

Executive Summary — Surrogate Model Validation Report

Use case: UCLOADS_1 | 03 September 2026 | 5 model(s)

REQUIREMENTS PARTIALLY MET

Recommended model: PyTorchNN

Avg R²: 0.9923 — Excellent

Q90 Accuracy — No output passes

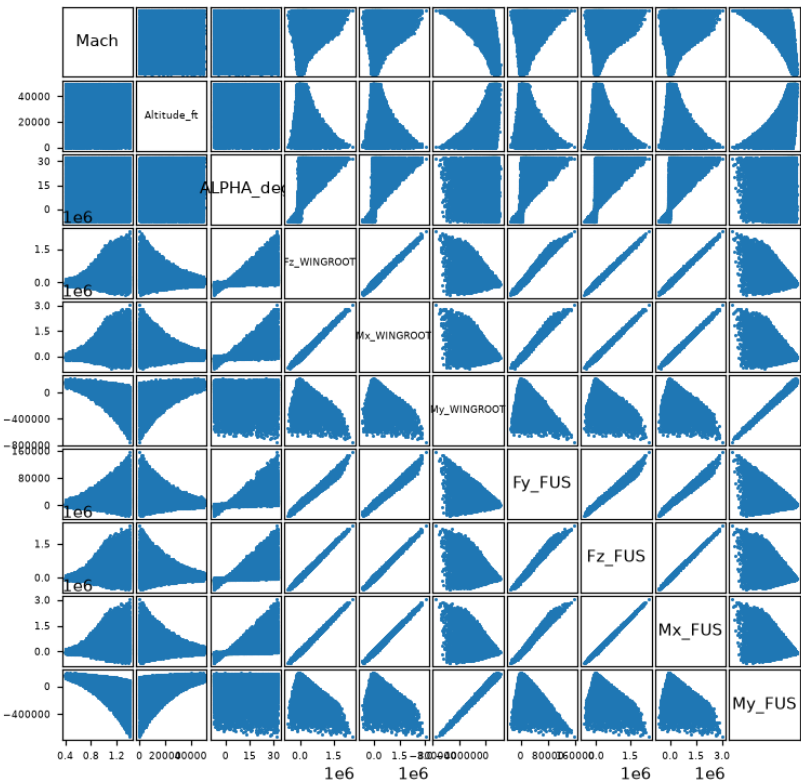
Fail: Fz_WINGROOT, Mx_WINGROOT, My_WINGROOT, Fy_FUS, Fz_FUS, Mx_FUS, My_FUS, Mz_FUS

Project Overview

Use case	UCLOADS_1
Inputs	Mach, Altitude_ft, ALPHA_deg
Outputs	8: Fz_WINGROOT, Mx_WINGROOT, My_WINGROOT, Fy_FUS, Fz_FUS, Mx_FUS, My_FUS, Mz_FUS
Train / Val / Test	70% / 10% / 20%
Accuracy target	Q90 < 10% relative error per output

Variable Correlation — Input vs Output

Variable Correlation



Data Statistics

Per-variable statistics over all data (train + val + test), as `Train_set.describe()` reports them in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum.

Inputs

	count	mean	std	min	P25	P50	P75	max
Mach	30,000	0.8993	0.2877	0.4	0.649	0.899	1.148	1.4
Altitude_ft	30,000	25041.4981	14450.3288	0.262	12538.2808	24908.6162	37629.5684	49999.7461
ALPHA_deg	30,000	12.6492	11.542	-7.499	2.7138	12.705	22.6283	32.5

Input variables — Train_set.describe()

Outputs

	count	mean	std	min	P25	P50	P75	max
Fz_WINGROOT	30,000	170703.0851	330033.1836	-552389.4375	-29380.1304	59228.2832	255270.5	2,310,000
Mx_WINGROOT	30,000	219817.2595	433912.0613	-719582.25	-41977.8672	72184.8438	328885.5859	3,050,000
My_WINGROOT	30,000	23493.4845	148166.7046	-754026.4375	-29575.6533	74652.375	127117.2227	216981.3906
Fy_FUS	30,000	13347.9878	18600.438	-30868.4082	2188.8232	7538.3909	18198.4678	156598.9844
Fz_FUS	30,000	170836.2941	329666.0959	-526906.0625	-28808.978	55768.1953	253443.3555	2,300,000
Mx_FUS	30,000	219365.6596	433865.0107	-683626.6875	-43647.6865	66522.2109	324146.1953	3,030,000
My_FUS	30,000	23581.2207	148229.718	-726381.75	-28888.1357	75282.5625	127646.9004	189144.7969
Mz_FUS	30,000	-22660.0748	27871.8468	-263049.875	-25834.228	-12596.9438	-7368.7069	7201.8691

Output variables — Train_set.describe()

Model Selection

Model	Avg R ²	Q90 pass (8)	KS pass (8)
MLP	0.9917	0/8	8/8
GradientBoosting	0.9907	0/8	7/8
RandomForest	0.9917	0/8	0/8
XGBoost	0.9590	0/8	7/8
PyTorchNN ★	0.9923	0/8	8/8

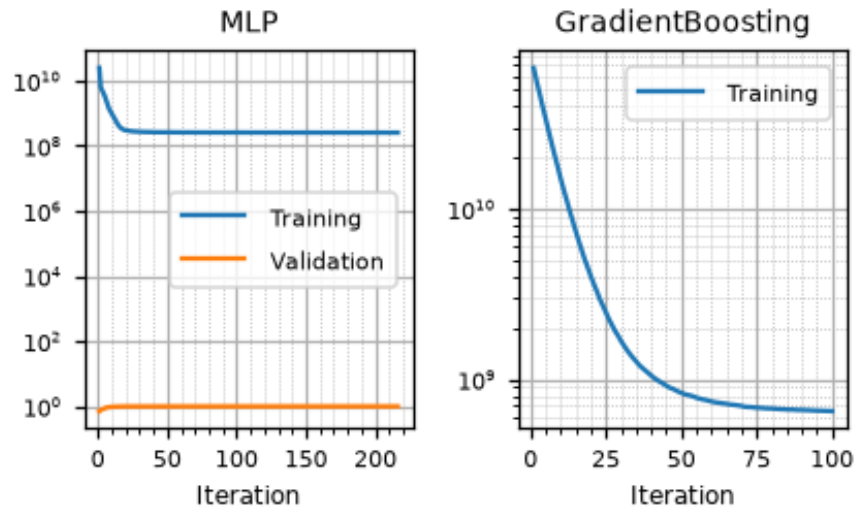
★ Recommended model. Q90 target: < 10%. KS: $p \geq 0.05$ = no overfitting.

Accuracy Assessment (Q90 & R²)

Output	MLP			GradientBoosting			RandomForest			XGBoost			PyTorchNN ★		
	R ²	Q90	Gap	R ²	Q90	Gap	R ²	Q90	Gap	R ²	Q90	Gap	R ²	Q90	Gap
Fz_WINGROOT	0.9900	0.2474	+0.1474	0.9880	0.2706	+0.1706	0.9894	0.2555	+0.1555	0.9452	0.5830	+0.4830	0.9901	0.2475	+0.1475
Mx_WINGROOT	0.9900	0.2501	+0.1501	0.9881	0.2733	+0.1733	0.9892	0.2596	+0.1596	0.9452	0.5863	+0.4863	0.9901	0.2494	+0.1494
My_WINGROOT	0.9900	0.2052	+0.1052	0.9896	0.2096	+0.1096	0.9892	0.2131	+0.1131	0.9898	0.2075	+0.1075	0.9901	0.2042	+0.1042
Fy_FUS	0.9876	0.2364	+0.1364	0.9879	0.2344	+0.1344	0.9896	0.2180	+0.1180	0.9462	0.4800	+0.3800	0.9901	0.2133	+0.1133
Fz_FUS	0.9975	0.1262	+0.0262	0.9954	0.1676	+0.0676	0.9972	0.1314	+0.0314	0.9523	0.5560	+0.4560	0.9975	0.1250	+0.0250
Mx_FUS	0.9975	0.1274	+0.0274	0.9954	0.1695	+0.0695	0.9973	0.1319	+0.0319	0.9523	0.5596	+0.4596	0.9975	0.1258	+0.0258
My_FUS	0.9974	0.1038	+0.0038	0.9970	0.1122	+0.0122	0.9969	0.1121	+0.0121	0.9972	0.1075	+0.0075	0.9975	0.1023	+0.0023
Mz_FUS	0.9836	0.2595	+0.1595	0.9840	0.2566	+0.1566	0.9851	0.2485	+0.1485	0.9437	0.4527	+0.3527	0.9859	0.2423	+0.1423

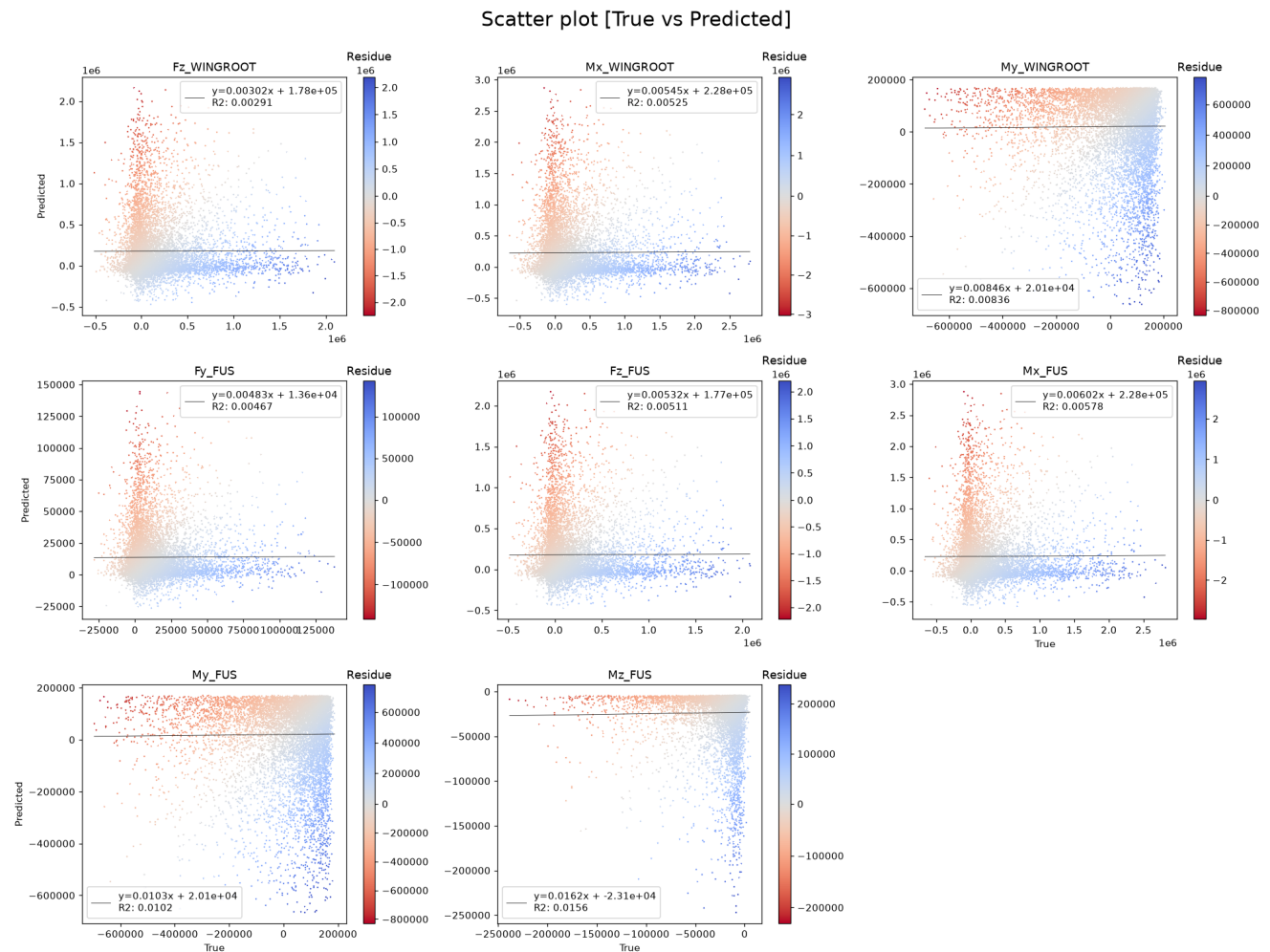
R²≥ 0.95 R²∈ [0.80, 0.95) R²< 0.80 or Q90 fail pass Gap=Q90−target (<0 ⇒ margin)

Training History



Training loss per iteration, log scale. A validation panel is shown for models trained with early stopping. Gradient boosting reports per-stage deviance averaged over its per-output estimators.

Predicted vs True — PyTorchNN



Data Split Quality

Metric	Value	Pass
Residual voxel proportion (≤ 0.05)	0.000	✓
Valid test proportion	0.966	
p-hacking test proportion (≤ 0.05 warn)	0.005	✓
Isolated test proportion (≤ 0.05 warn)	0.029	✓
Chi ² p-value (≥ 0.05)	—	—

No p-hacking concern: only 0.5% of test points lie closer to a training point than training points are to each other, so the test error is not flattered.

Overfitting Check (KS Test) — PyTorchNN

Output	KS p-value
Fz_WINGROOT	0.891
Mx_WINGROOT	0.172
My_WINGROOT	0.663
Fy_FUS	0.586
Fz_FUS	0.501
Mx_FUS	0.523
My_FUS	0.791
Mz_FUS	0.479

■ $p \geq 0.05$ — no overfitting ■ $p < 0.05$ — overfitting detected

Deep Analysis

UCLOADS_1

Winner Model — Full Validation

Deep Analysis — PyTorchNN

Mirrors `validation_template.ipynb` — all computations from validation CSVs

- 3.1 Data Overview
- 3.2 Train-Test Split Analysis
- 3.3 Error Quantification — $P(E)$
- 3.4 $P(E|X)$ — Error Conditional on Inputs
- 3.5 $P(E|Y)$ — Error Conditional on Outputs
- 3.6 Uncertainty Analysis

Data Overview

Statistical summary of inputs and outputs over all data (train + val + test), as reported by `describe()` in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum per variable. All figures in this part are drawn by `validationlib`, matching the extended validation notebook.

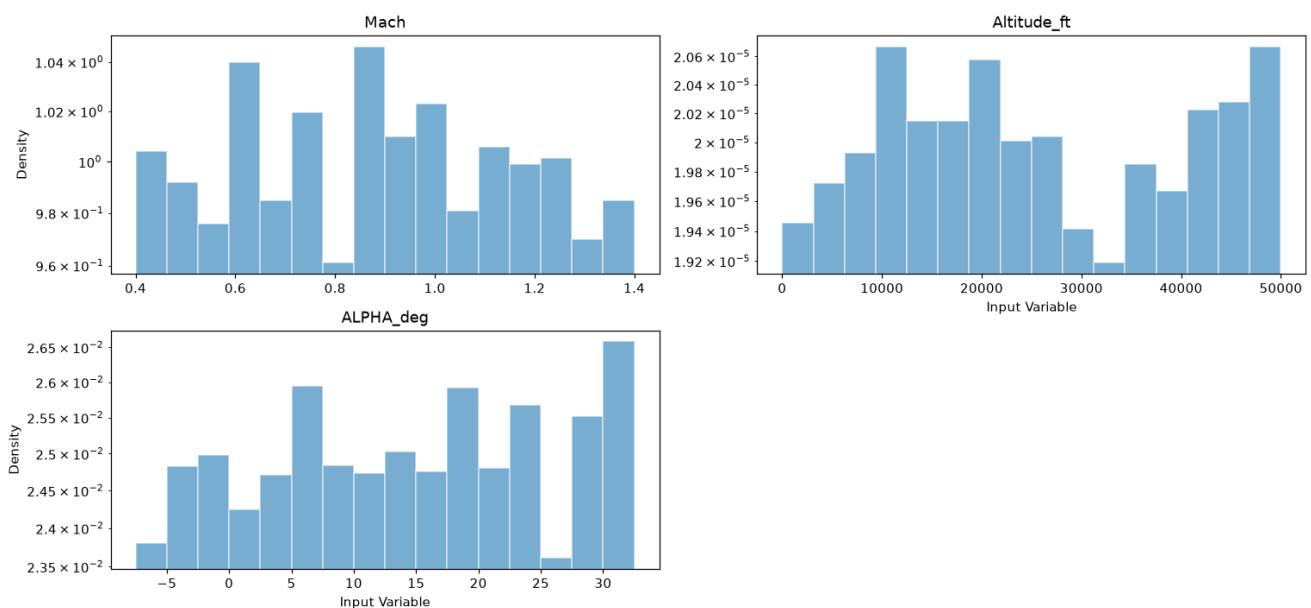
Input Statistics

	count	mean	std	min	P25	P50	P75	max
Mach	30,000	0.8993	0.2877	0.4	0.649	0.899	1.148	1.4
Altitude_ft	30,000	25041.4981	14450.3288	0.262	12538.2808	24908.6162	37629.5684	49999.7461
ALPHA_deg	30,000	12.6492	11.542	-7.499	2.7138	12.705	22.6283	32.5

Input variables — `Train_set.describe()`

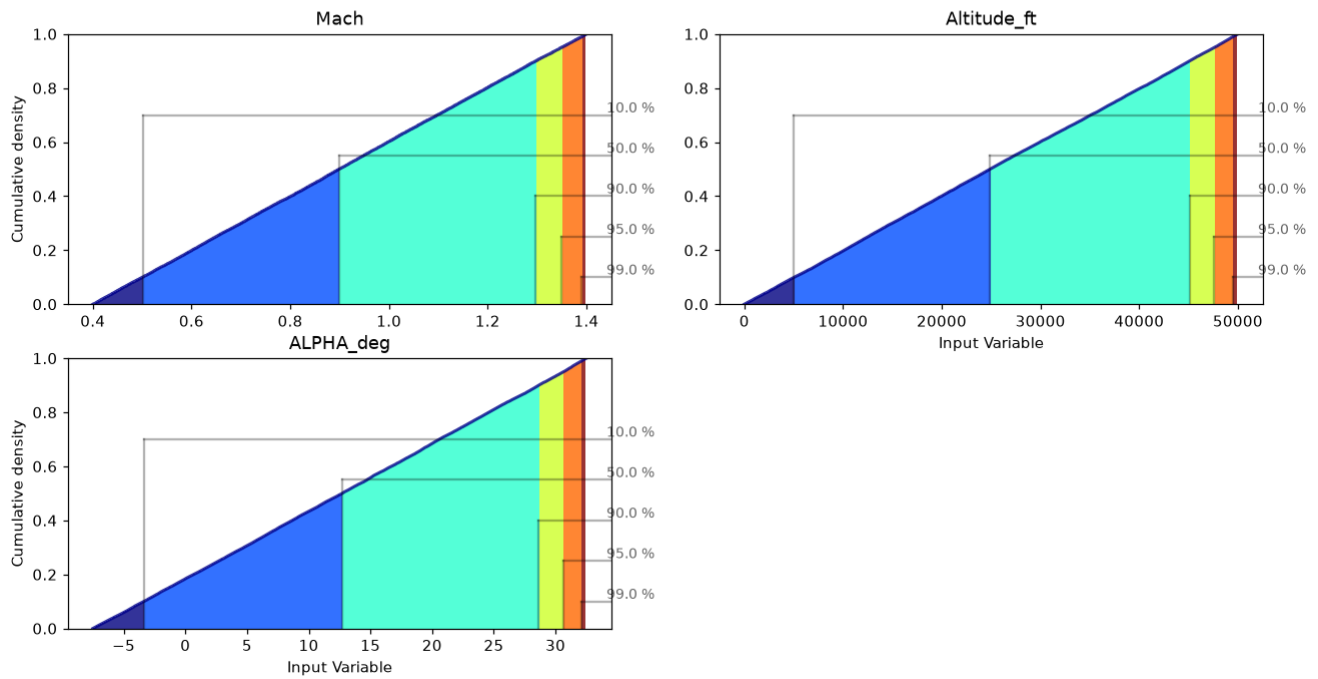
Input Histograms

Histogram of [Input Variable] inside $\mu \pm 3\sigma$



Input Cumulative Distributions

Cumulative plot of [Input Variable]

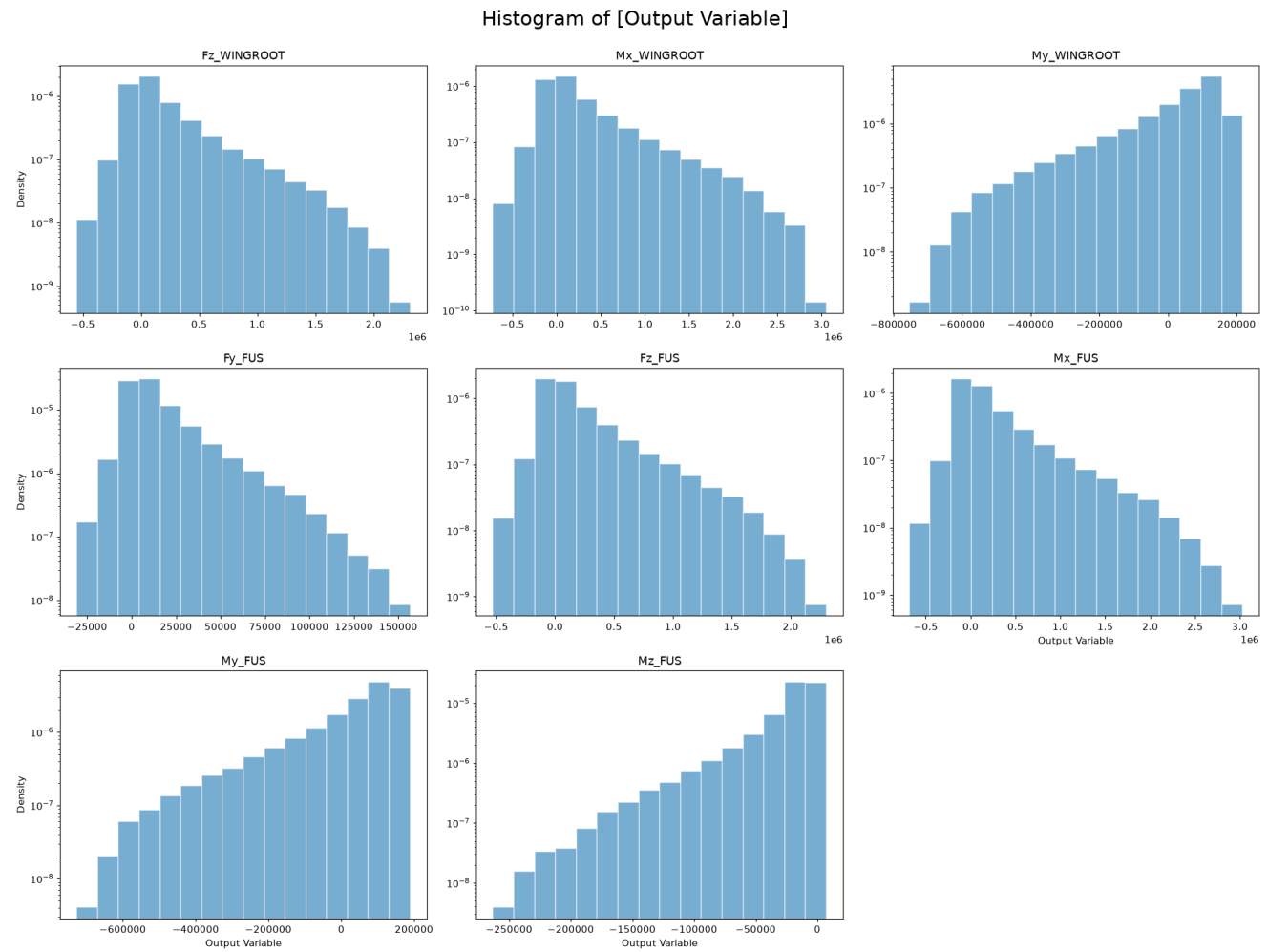


Output Statistics

	count	mean	std	min	P25	P50	P75	max
Fz_WINGROOT	30,000	170703.0851	330033.1836	-552389.4375	-29380.1304	59228.2832	255270.5	2,310,000
Mx_WINGROOT	30,000	219817.2595	433912.0613	-719582.25	-41977.8672	72184.8438	328885.5859	3,050,000
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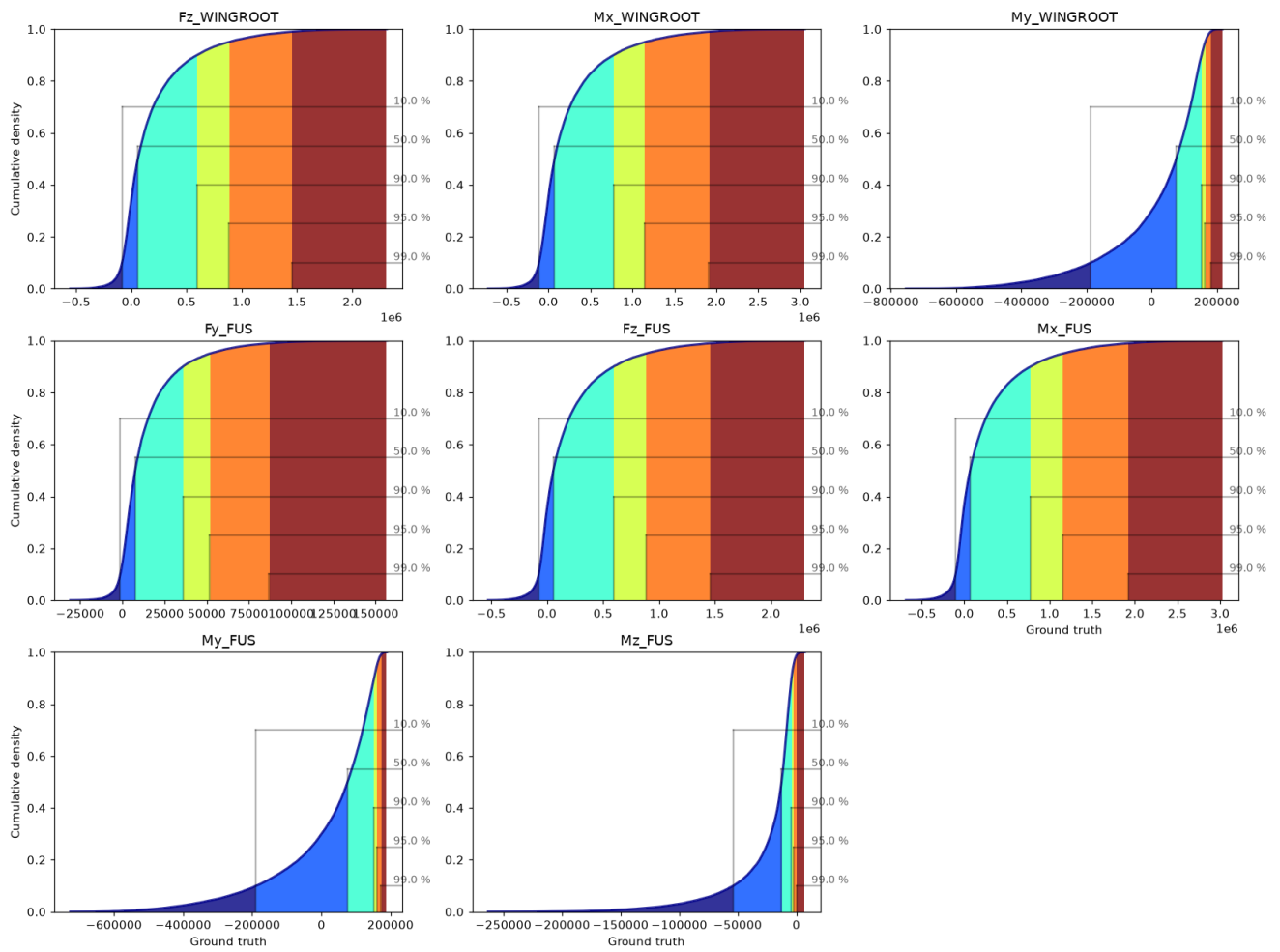
Output variables — Train_set.describe()

Output Histograms



Output Cumulative Distributions

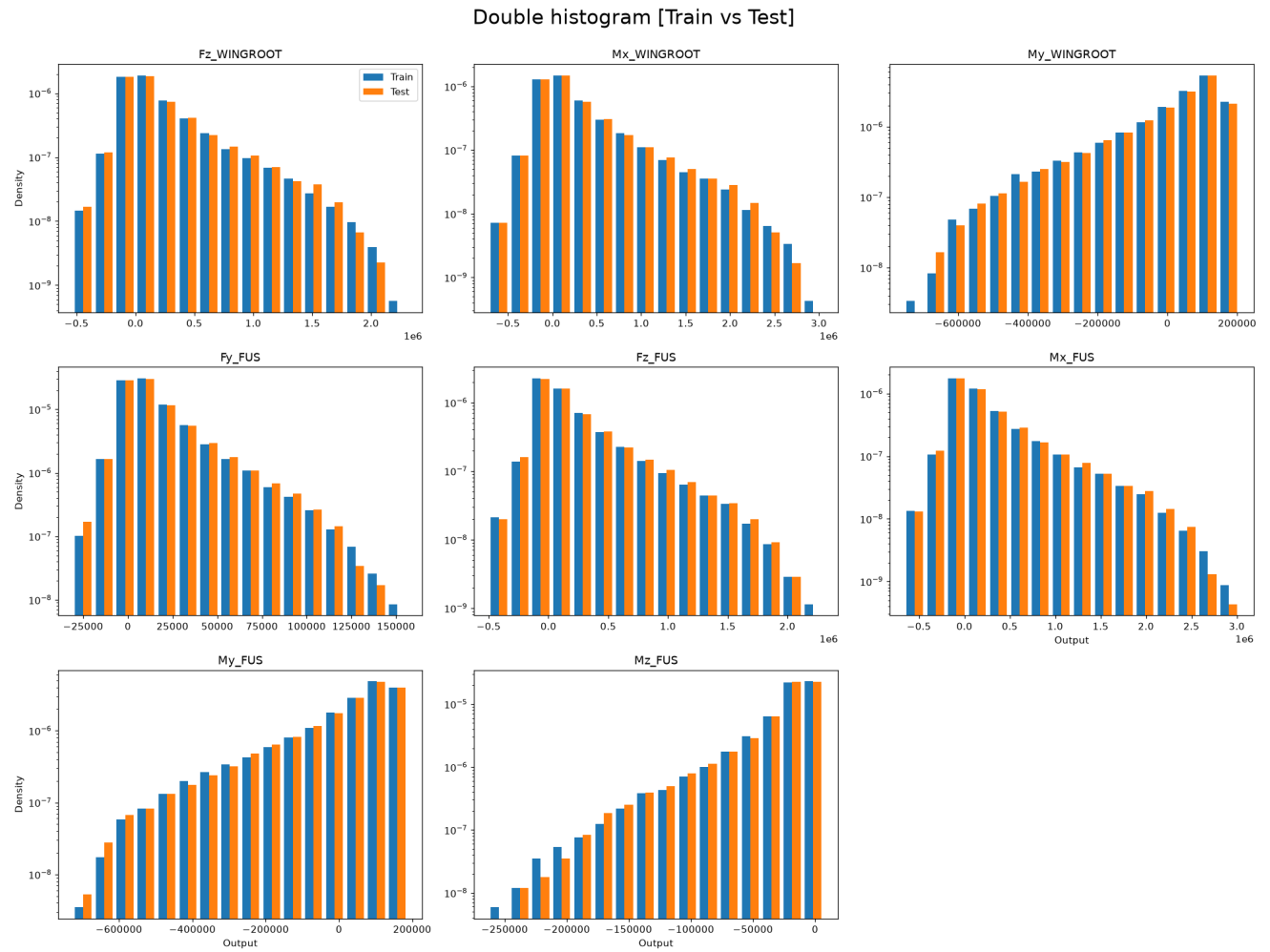
Cumulative plot of [Ground truth]



Train-Test Split Analysis

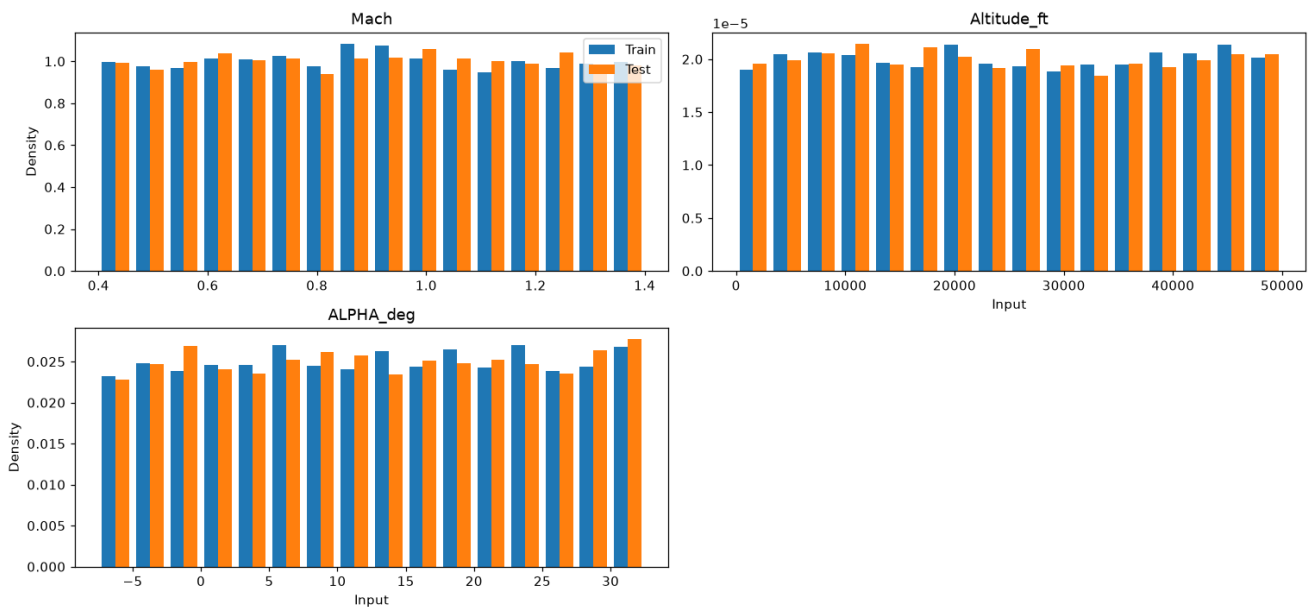
A well-designed split produces training and test distributions that are similar but not identical. The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test are used to assess distributional similarity. H_0 : both sets come from the same distribution. $p < 0.05$ rejects H_0 (undesirable if too low; undesirable if the split was done by stratification and p is too high).

Output Distributions: Train vs Test



Input Distributions: Train vs Test

Double histogram [Train vs Test]



KS and AD Test Results

	KS	AD
Fz_WINGROOT	0.6637	0.25
Mx_WINGROOT	0.8629	0.25
My_WINGROOT	0.7802	0.25
Fy_FUS	0.8335	0.25
Fz_FUS	0.8534	0.25
Mx_FUS	0.9062	0.25
My_FUS	0.4894	0.25
Mz_FUS	0.7912	0.25

Train vs test p-values. Red: $p < 0.05$ (distributions differ).

Split Quality — Voxel Tessellation Proximity (p-hacking check)

Beyond matching distributions, a split can still flatter the model if test points sit right next to training points (p-hacking) or probe regions the model never saw (isolated, residual voxel). The VTPM method from `validationlib` (SF_9 cell 9.0) classifies every test point.

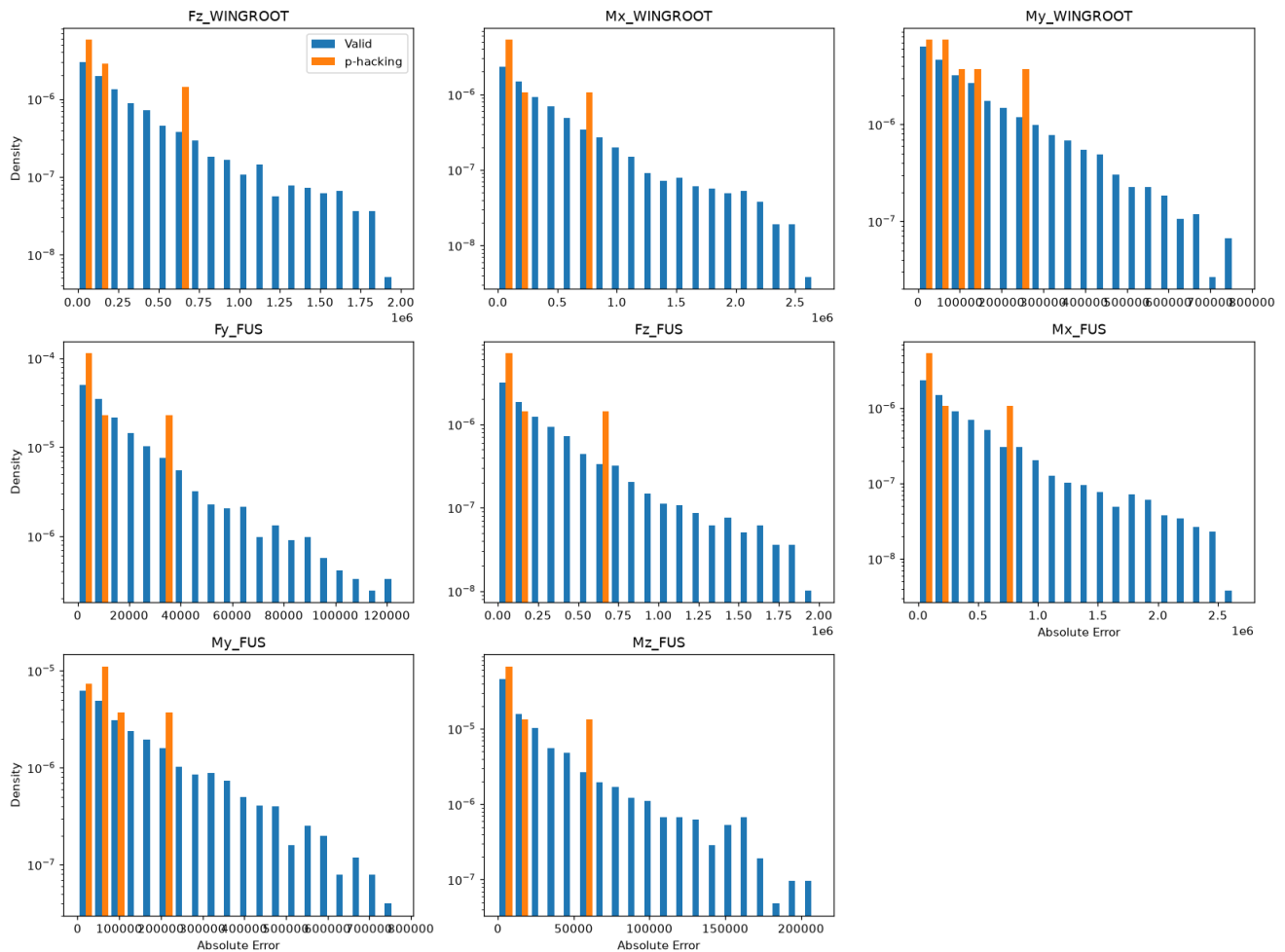
Metric	Value	Pass
Residual voxel proportion (≤ 0.05)	0.000	✓
Valid test proportion	0.966	
p-hacking test proportion (≤ 0.05 warn)	0.005	✓
Isolated test proportion (≤ 0.05 warn)	0.029	✓
Chi ² p-value (≥ 0.05)	—	—

No p-hacking concern: only 0.5% of test points lie closer to a training point than training points are to each other, so the test error is not flattered.

Error Distribution: Valid vs p-hacking Test Points

Absolute error of the test points flagged as p-hacking against the valid ones, exactly as the extended validation notebook draws it. Matching distributions (high p-values) mean the flagged points do not distort the reported error.

Double histogram [Valid vs p-hacking]



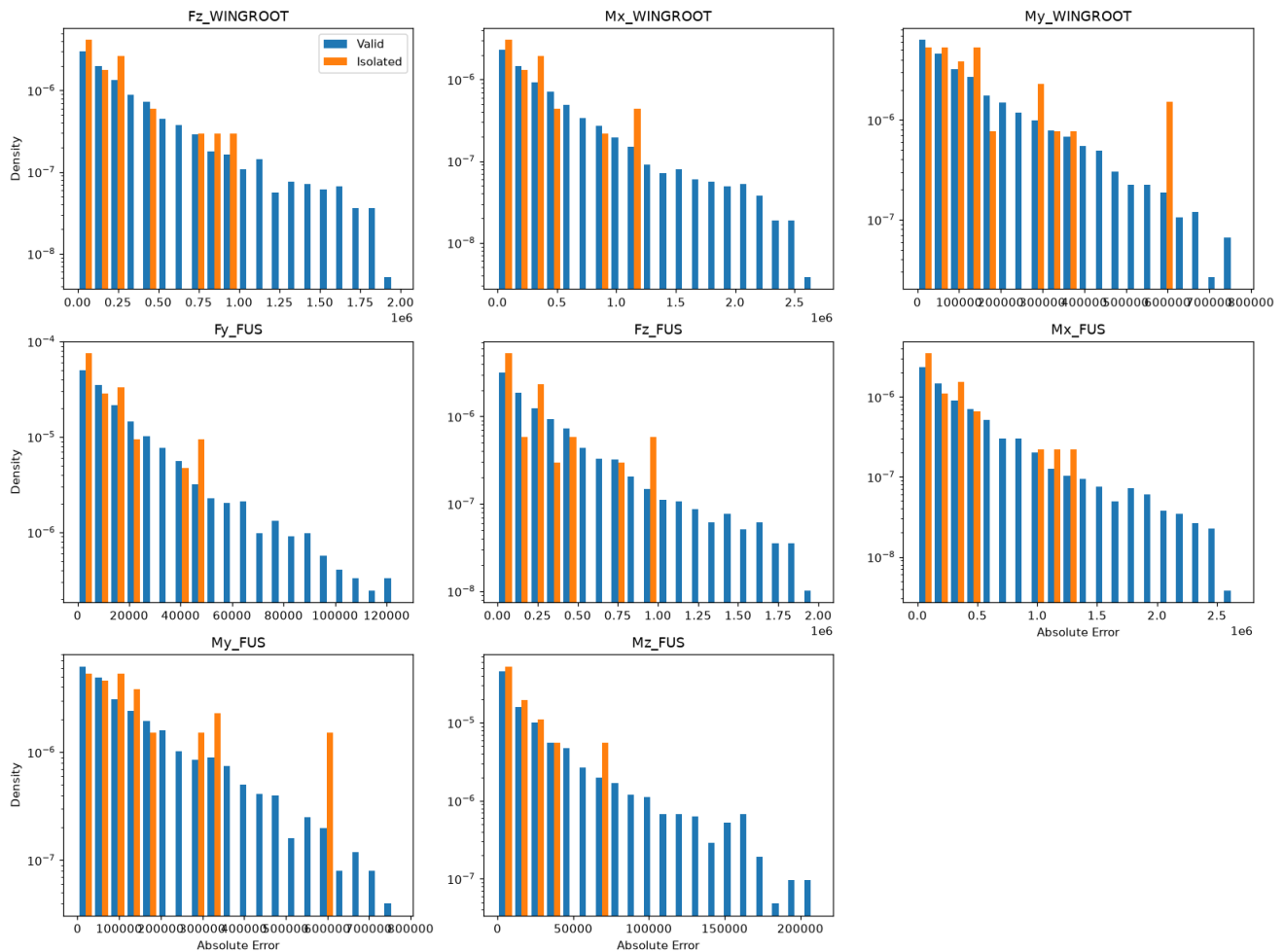
	AD	KS	KW
Fz_WINGROOT	0.25	0.1981	0.1892
Mx_WINGROOT	0.0324	0.0764	0.0342
My_WINGROOT	0.25	0.4799	0.23
Fy_FUS	0.1025	0.0688	0.0688
Fz_FUS	0.0124	0.0221	0.02
Mx_FUS	0.0166	0.0688	0.0295
My_FUS	0.25	0.3994	0.2023
Mz_FUS	0.0324	0.0213	0.0449

Valid ($n = 1959$) vs p-hacking ($n = 7$) absolute-error distributions, on a subsample of the test set. Red: $p < 0.05$ (the flagged points behave differently).

Error Distribution: Valid vs Isolated Test Points

Same comparison for the test points with no training neighbour.

Double histogram [Valid vs Isolated]



	AD	KS	KW
Fz_WINGROOT	0.1027	0.036	0.0657
Mx_WINGROOT	0.0609	0.0494	0.0455
My_WINGROOT	0.25	0.6467	0.9086
Fy_FUS	0.0744	0.0607	0.0589
Fz_FUS	0.0636	0.0778	0.0459
Mx_FUS	0.0836	0.0829	0.0561
My_FUS	0.25	0.574	0.9619
Mz_FUS	0.25	0.4272	0.3739

Valid ($n = 1959$) vs Isolated ($n = 34$) absolute-error distributions, on a subsample of the test set. Red: $p < 0.05$ (the flagged points behave differently).

Error Quantification — P(E)

Two error metrics are studied:

- Residue: $e_i = y_i - \hat{y}_i$. A centred ($\mu \approx 0$), symmetric distribution indicates an unbiased model.
- Absolute Error: $|e_i|$. Always non-negative; its Q90 must satisfy the accuracy requirement.

Residue Statistics Table

Bootstrap confidence bounds shown as superscript (upper) and subscript (lower).

	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
Fz_WINGROOT	10,000	-2,250,000	2,190,000	-4,860	⁵¹⁵⁰ ₋₁₆₁₀₀	-1,120	⁵⁷³⁰ ₋₆₆₄₀	478,000	⁴⁸⁹⁰⁰⁰ ₄₆₉₀₀₀	395,000	⁴⁰⁹⁰⁰⁰ ₃₈₄₀₀₀	2.53	^{2.7} _{2.32}	-0.104	^{-0.0052} _{-0.193}
Mx_WINGROOT	10,000	-3,030,000	2,920,000	-7,070	⁴¹⁹⁰ ₋₁₉₄₀₀	-885	⁷⁷¹⁰ ₋₇₈₄₀	628,000	⁶⁴⁴⁰⁰⁰ ₆₁₅₀₀₀	516,000	⁵³⁶⁰⁰⁰ ₄₉₉₀₀₀	2.61	^{2.8} _{2.42}	-0.112	^{-0.0096} _{-0.204}
My_WINGROOT	10,000	-834,000	780,000	2,170	⁶²⁴⁰ ₋₂₄₅₀	2,370	⁵⁰⁷⁰ ₋₃₈₄	210,000	²¹⁴⁰⁰⁰ ₂₀₇₀₀₀	201,000	²⁰⁷⁰⁰⁰ ₁₉₆₀₀₀	1.39	^{1.53} _{1.24}	-0.0099	^{0.0654} _{-0.0697}
Fy_FUS	10,000	-141,000	143,000	-244	²⁵¹ ₋₈₈₀	-66.1	²⁷¹ ₋₄₃₀	26,800	²⁷⁴⁰⁰ ₂₆₂₀₀	22,400	²³¹⁰⁰ ₂₁₇₀₀	3.17	^{3.44} _{2.77}	-0.114	^{0.0016} _{-0.235}
Fz_FUS	10,000	-2,220,000	2,200,000	-4,950	⁴¹⁸⁰ ₋₁₂₃₀₀	213	⁴⁹²⁰ ₋₃₈₉₀	478,000	⁴⁸⁶⁰⁰⁰ ₄₇₁₀₀₀	395,000	⁴⁰⁸⁰⁰⁰ ₃₈₃₀₀₀	2.55	^{2.74} _{2.33}	-0.103	^{0.011} _{-0.191}
Mx_FUS	10,000	-2,980,000	2,940,000	-6,810	⁴⁷³⁰ ₋₁₈₁₀₀	489	⁶⁸⁴⁰ ₋₄₆₇₀	629,000	⁶⁴³⁰⁰⁰ ₆₁₄₀₀₀	513,000	⁵²⁷⁰⁰⁰ ₅₀₁₀₀₀	2.63	^{2.81} _{2.39}	-0.105	^{-0.0283} _{-0.209}
My_FUS	10,000	-828,000	780,000	2,340	⁸¹¹⁰ ₋₉₁₁	2,520	⁵⁵⁹⁰ _{-90.5}	210,000	²¹³⁰⁰⁰ ₂₀₆₀₀₀	200,000	²⁰⁴⁰⁰⁰ ₁₉₅₀₀₀	1.41	^{1.52} _{1.29}	-0.0125	^{0.0684} _{-0.08}
Mz_FUS	10,000	-230,000	238,000	469	¹²²⁰ ₋₄₀₆	26.8	⁴²³ ₋₂₉₀	40,400	⁴¹⁷⁰⁰ ₃₉₃₀₀	23,000	²³⁸⁰⁰ ₂₂₃₀₀	5.22	^{5.97} _{4.75}	0.188	^{0.399} _{0.0363}

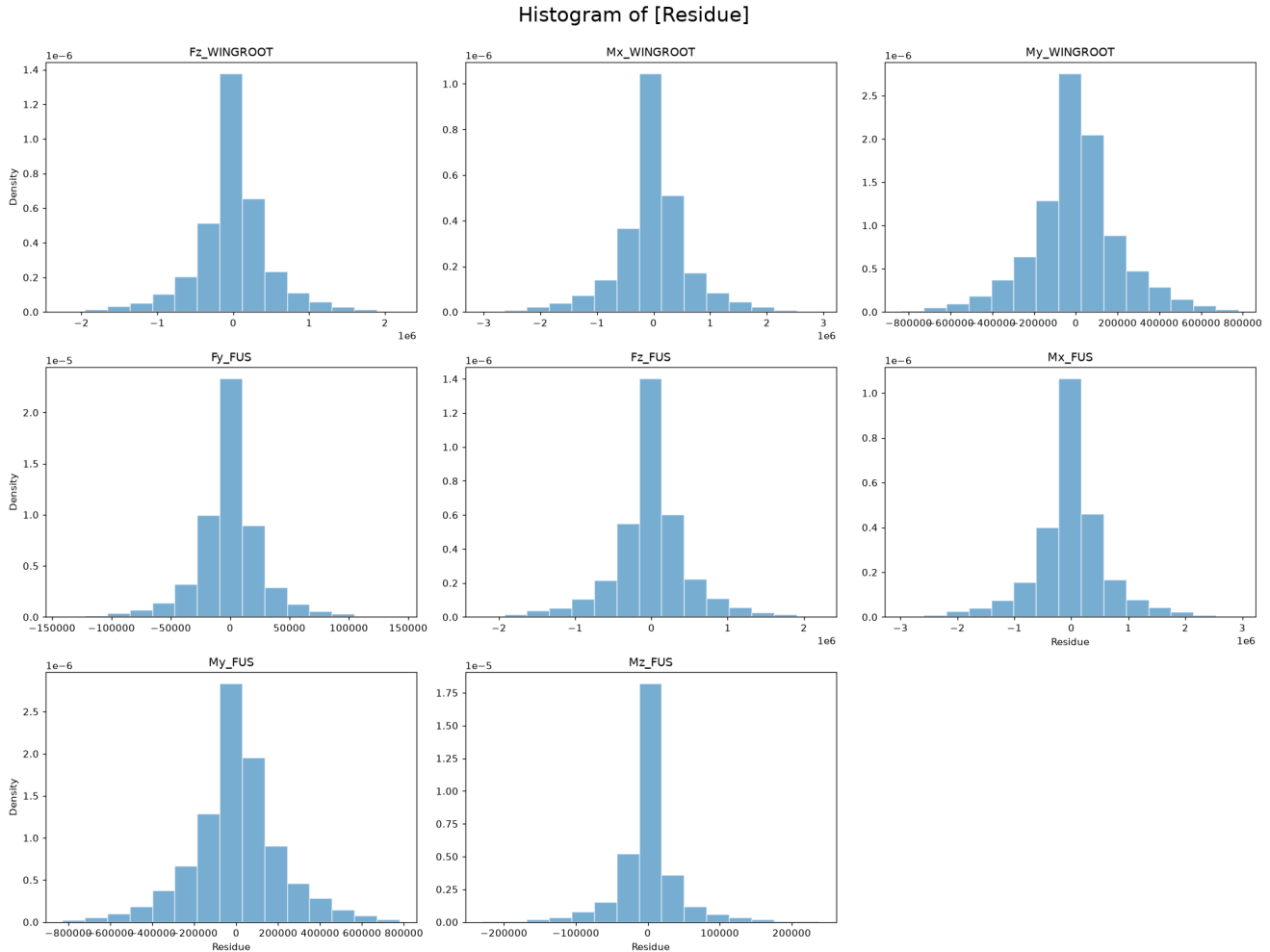
Residue statistics with bootstrap confidence intervals (superscript/subscript bounds).

Absolute Error Statistics Table

	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
Fz_WINGROOT	10,000	42.6	2,250,000	325,000	³³¹⁰⁰⁰ ₃₁₇₀₀₀	198,000	²⁰⁴⁰⁰⁰ ₁₉₁₀₀₀	351,000	³⁵⁹⁰⁰⁰ ₃₄₂₀₀₀	362,000	³⁷⁶⁰⁰⁰ ₃₅₀₀₀₀	3.44	^{3.79} _{3.11}	1.82	^{1.87} _{1.76}
Mx_WINGROOT	10,000	1.58	3,030,000	424,000	⁴³⁴⁰⁰⁰ ₄₁₆₀₀₀	258,000	²⁶⁴⁰⁰⁰ ₂₄₉₀₀₀	463,000	⁴⁷⁶⁰⁰⁰ ₄₅₂₀₀₀	472,000	⁴⁸⁶⁰⁰⁰ ₄₅₇₀₀₀	3.53	^{3.84} _{3.18}	1.84	^{1.89} _{1.78}
My_WINGROOT	10,000	26.1	834,000	150,000	¹⁵³⁰⁰⁰ ₁₄₈₀₀₀	100,000	¹⁰³⁰⁰⁰ ₉₇₀₀₀	147,000	¹⁵⁰⁰⁰⁰ ₁₄₄₀₀₀	175,000	¹⁸²⁰⁰⁰ ₁₇₁₀₀₀	1.98	^{2.2} _{1.72}	1.48	^{1.52} _{1.43}
Fy_FUS	10,000	1.61	143,000	18,100	¹⁸⁶⁰⁰ ₁₇₈₀₀	11,200	¹¹⁶⁰⁰ ₁₀₉₀₀	19,800	²⁰²⁰⁰ ₁₉₃₀₀	19,400	¹⁹⁹⁰⁰ ₁₈₇₀₀	4.79	^{5.27} _{4.3}	2.02	^{2.09} _{1.95}
Fz_FUS	10,000	12	2,220,000	324,000	³³¹⁰⁰⁰ ₃₁₇₀₀₀	197,000	²⁰⁴⁰⁰⁰ ₁₉₂₀₀₀	352,000	³⁶⁰⁰⁰⁰ ₃₄₅₀₀₀	364,000	³⁷⁷⁰⁰⁰ ₃₅₃₀₀₀	3.43	^{3.72} _{3.12}	1.82	^{1.87} _{1.77}
Mx_FUS	10,000	26.5	2,980,000	424,000	⁴³²⁰⁰⁰ ₄₁₄₀₀₀	256,000	²⁶²⁰⁰⁰ ₂₄₉₀₀₀	465,000	⁴⁷³⁰⁰⁰ ₄₅₃₀₀₀	476,000	⁴⁹⁰⁰⁰⁰ ₄₅₇₀₀₀	3.52	^{3.84} _{3.16}	1.84	^{1.9} _{1.77}
My_FUS	10,000	12.6	828,000	150,000	¹⁵³⁰⁰⁰ ₁₄₈₀₀₀	99,800	¹⁰³⁰⁰⁰ ₉₆₉₀₀	147,000	¹⁵⁰⁰⁰⁰ ₁₄₄₀₀₀	175,000	¹⁸¹⁰⁰⁰ ₁₇₁₀₀₀	1.98	^{2.2} _{1.75}	1.48	^{1.53} _{1.43}
Mz_FUS	10,000	0.821	238,000	24,400	²⁴⁹⁰⁰ ₂₃₈₀₀	11,500	¹¹⁹⁰⁰ ₁₁₂₀₀	32,200	³²⁹⁰⁰ ₃₁₂₀₀	26,100	²⁷³⁰⁰ ₂₄₈₀₀	6.42	^{6.99} _{5.86}	2.37	^{2.45} _{2.28}

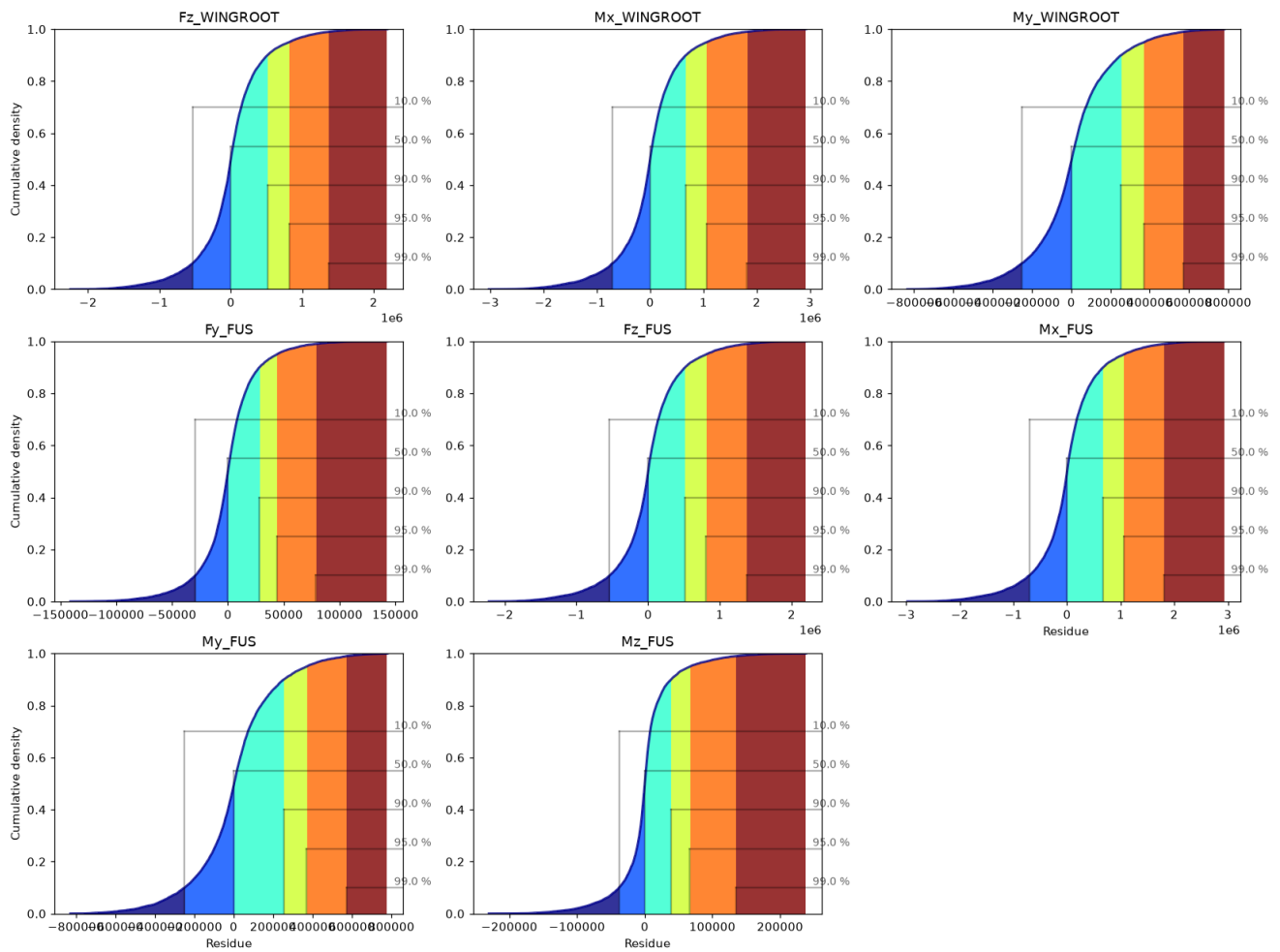
Absolute-error statistics with bootstrap confidence intervals.

Residue Histograms



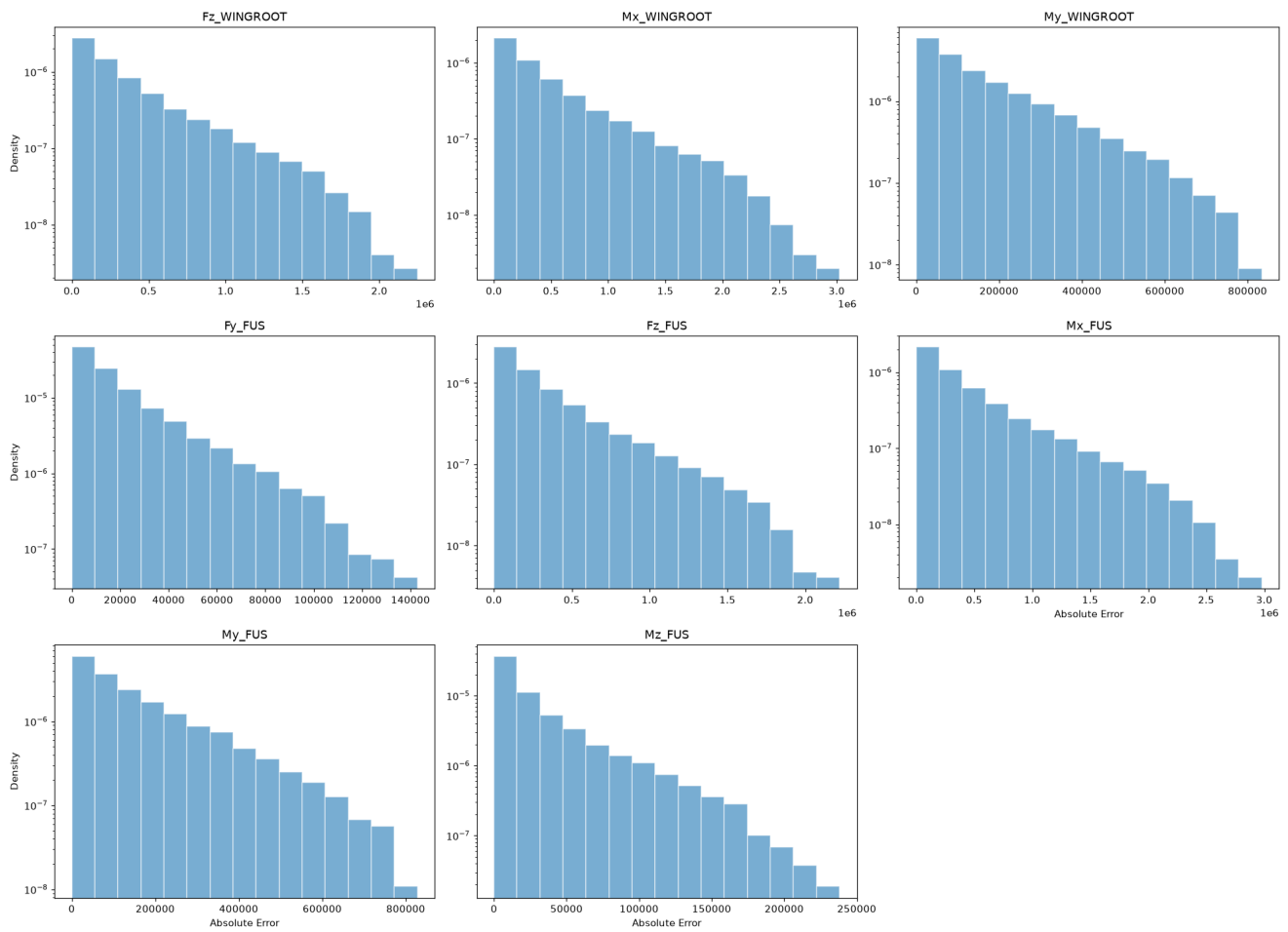
Residue CDF

Cumulative plot of [Residue]



Absolute Error Histograms

Histogram of [Absolute Error]



Q90 Accuracy Requirement

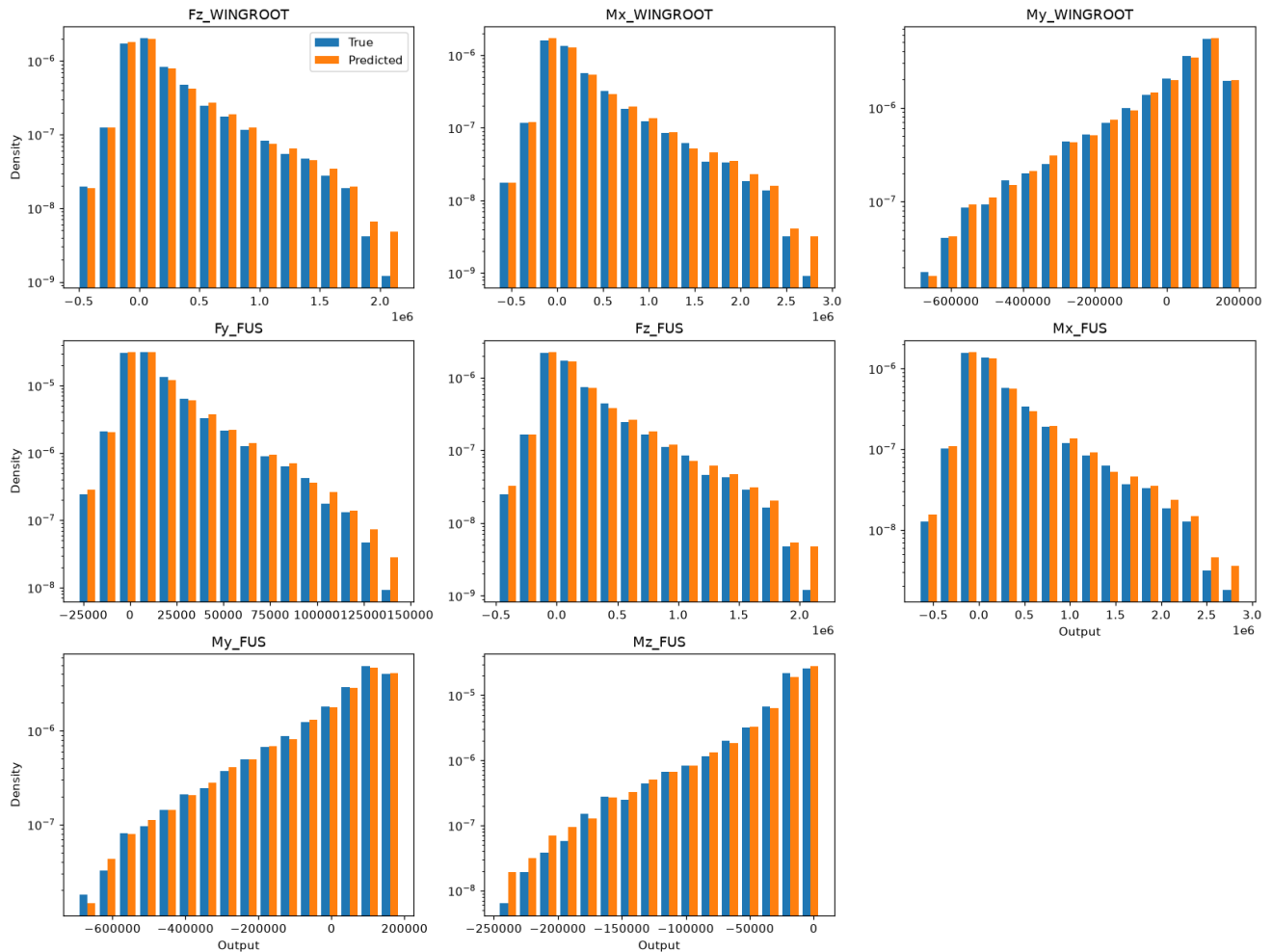
Output	Q90	Target	Status
Fz_WINGROOT	12.8856	10%	FAIL
Mx_WINGROOT	12.8081	10%	FAIL
My_WINGROOT	4.9689	10%	FAIL
Fy_FUS	10.8875	10%	FAIL
Fz_FUS	13.1123	10%	FAIL
Mx_FUS	13.3095	10%	FAIL
My_FUS	4.8529	10%	FAIL
Mz_FUS	5.7248	10%	FAIL

■ $p \geq 0.05$ — pass
 ■ $p < 0.05$ — fail

True vs Predicted Distribution Comparison

AD and KS tests comparing the distribution of true vs. predicted values on the test set. A significant difference indicates the model is not reproducing the output distribution faithfully.

Double histogram [True vs Predicted]

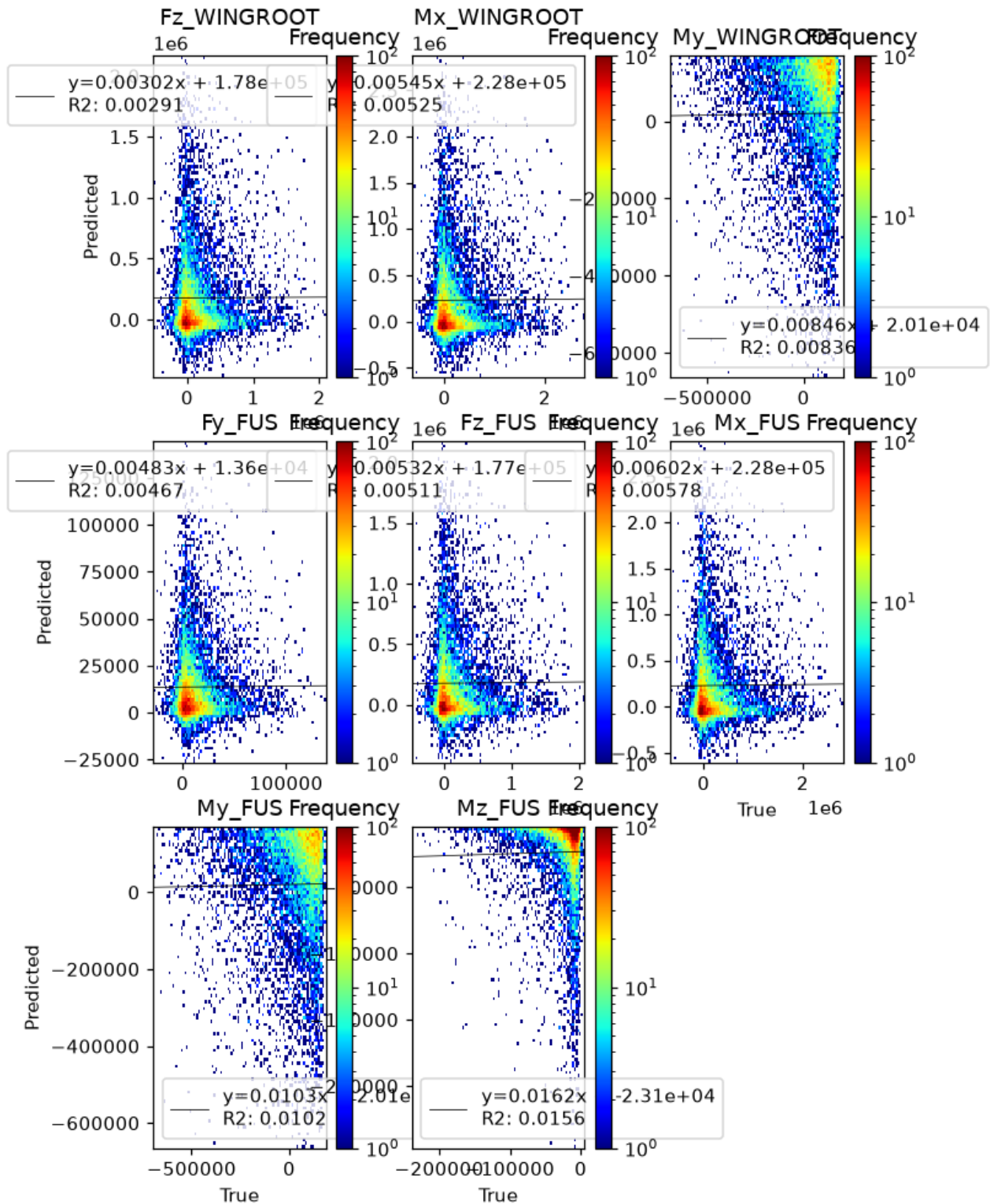


	AD	KS
Fz_WINGROOT	0.0043	0.0022
Mx_WINGROOT	0.0012	0.000529
My_WINGROOT	0.001	7.13e-05
Fy_FUS	0.0337	0.0397
Fz_FUS	0.231	0.251
Mx_FUS	0.25	0.4463
My_FUS	0.0355	0.2169
Mz_FUS	0.001	1.33e-55

True vs predicted distribution tests. Red: $p < 0.05$.

True vs Predicted — 2D Histogram

Log-scaled frequency with a linear fit; the normalized slope should be close to 1.



P(E|X) — Error Conditional on Inputs

We study whether the model error depends on the input variables. Two approaches are used: (1) a Kruskal-Wallis (KW) test on binned inputs (Sturges rule for bin count), and (2) violin plots showing the error distribution within each input bin.

Significant input-conditional bias ($p < 0.05$) detected for: **Mach→Fz_WINGROOT, Mach→Mx_WINGROOT, Mach→My_WINGROOT, Mach→Fy_FUS, Mach→Fz_FUS, Mach→Mx_FUS, Mach→My_FUS, Mach→Mz_FUS, Altitude_ft→Fz_WINGROOT, Altitude_ft→Mx_WINGROOT....** These inputs explain residual variance — consider polynomial features or interaction terms.

Bias Detection Table (KW p-values)

Green ($p \geq 0.05$) = residue uniform across input bins; red ($p < 0.05$) = significant bias.

	Fz_WINGROOT	Mx_WINGROOT	My_WINGROOT	Fy_FUS	Fz_FUS	Mx_FUS	My_FUS	Mz_FUS
Mach_binned	4.09e-156	1.83e-155	0	3.85e-121	2.51e-156	1.99e-158	0	2.66e-175
Altitude_ft_binned	8.46e-113	2.89e-110	0	3.66e-120	7.38e-114	1.36e-112	0	2.62e-146
ALPHA_deg_binned	0	0	0.0151	0	0	0	0.0233	0

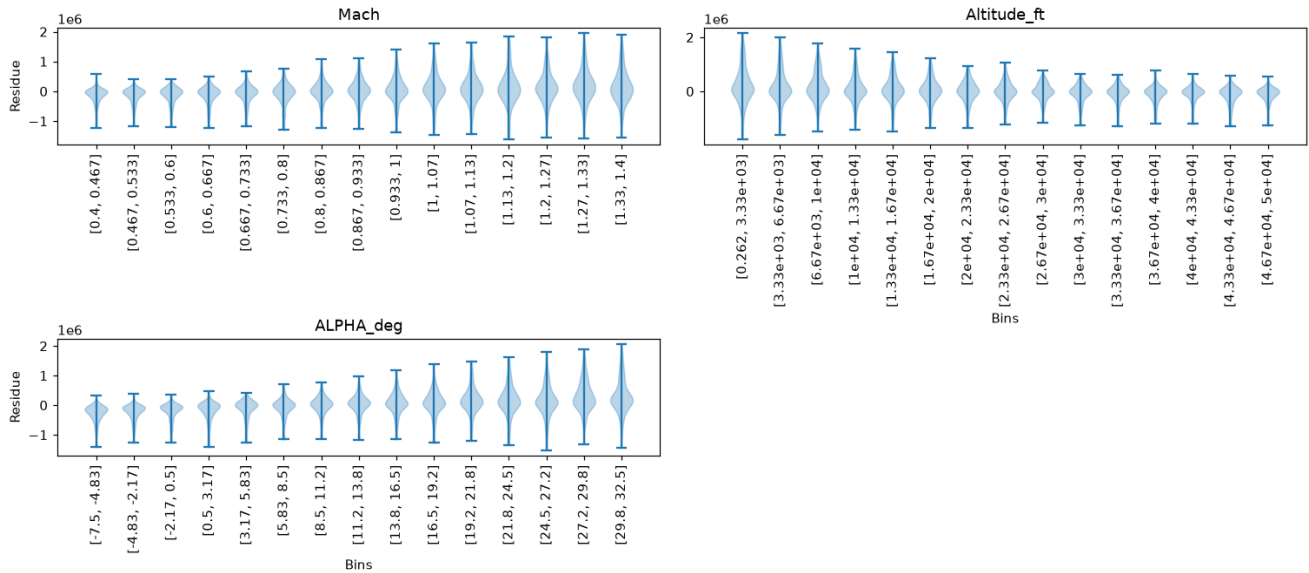
Kruskal-Wallis p-values on Sturges-binned inputs. Red: $p < 0.05$ (error depends on that input).

Residue Violin Plots per Output (binned by input)

Each panel shows the residue distribution for each input bin. A flat median line (dashed at 0) and symmetric violins indicate absence of bias.

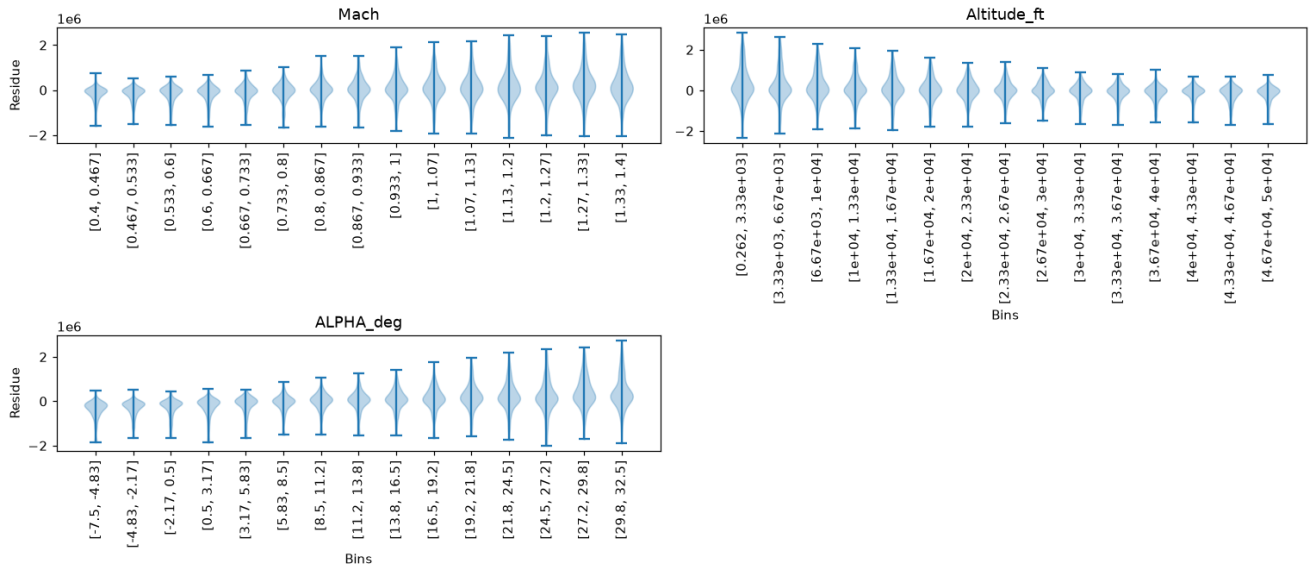
Residue — Fz_WINGROOT

Fz_WINGROOT
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$



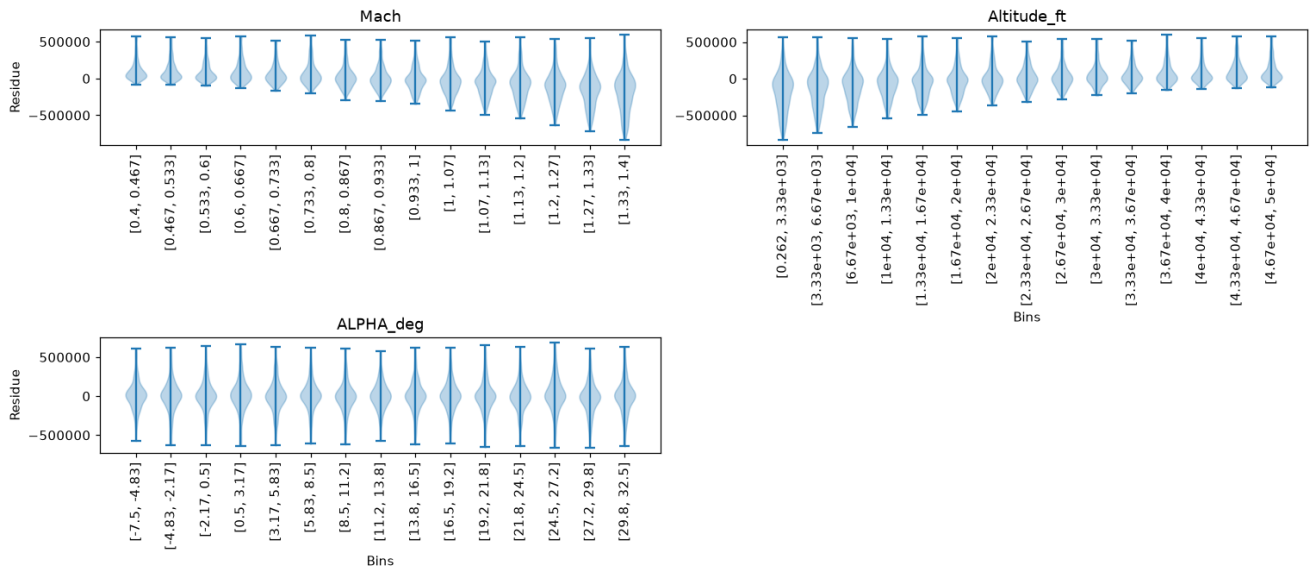
Residue — Mx_WINGROOT

Mx_WINGROOT
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$



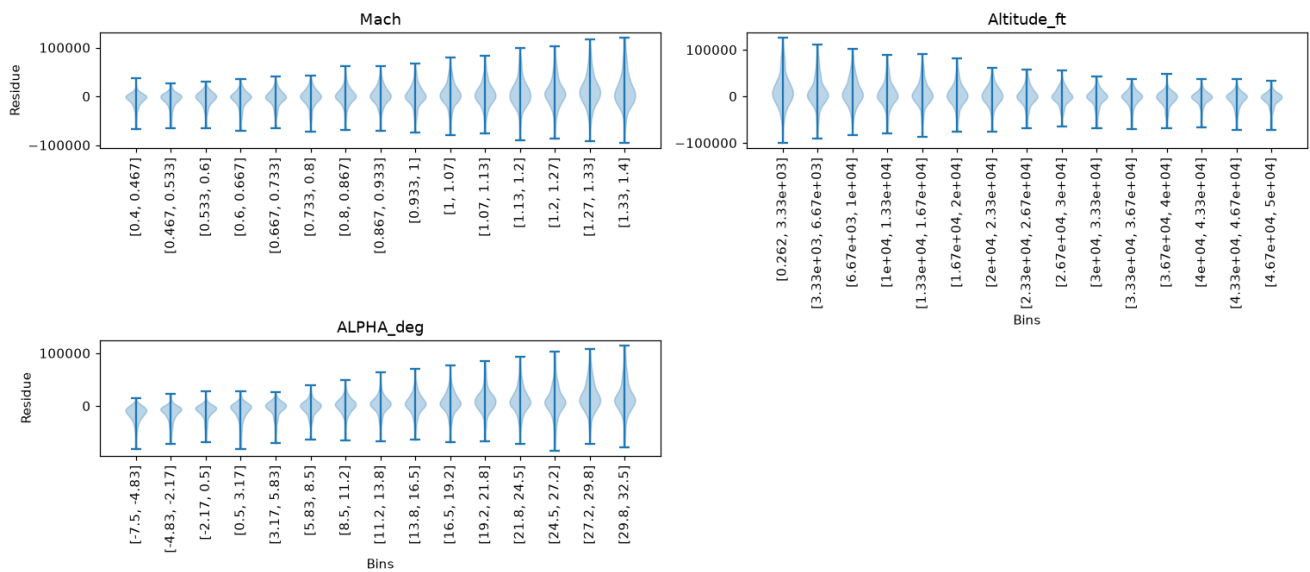
Residue — My_WINGROOT

My_WINGROOT
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$



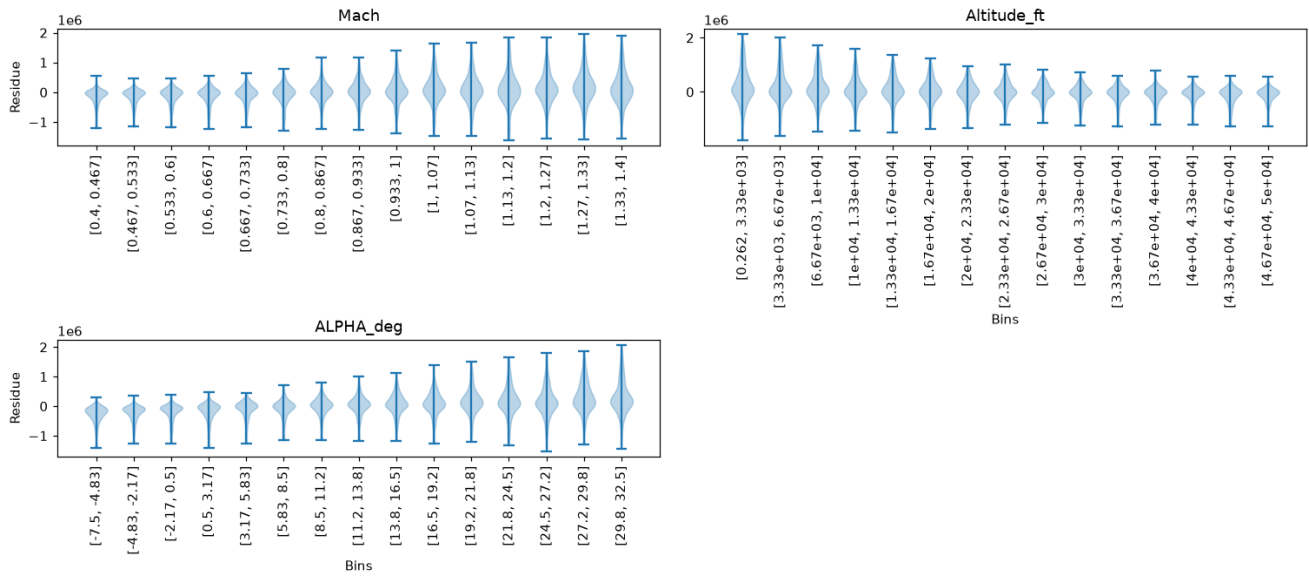
Residue — Fy_FUS

Fy_FUS
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$



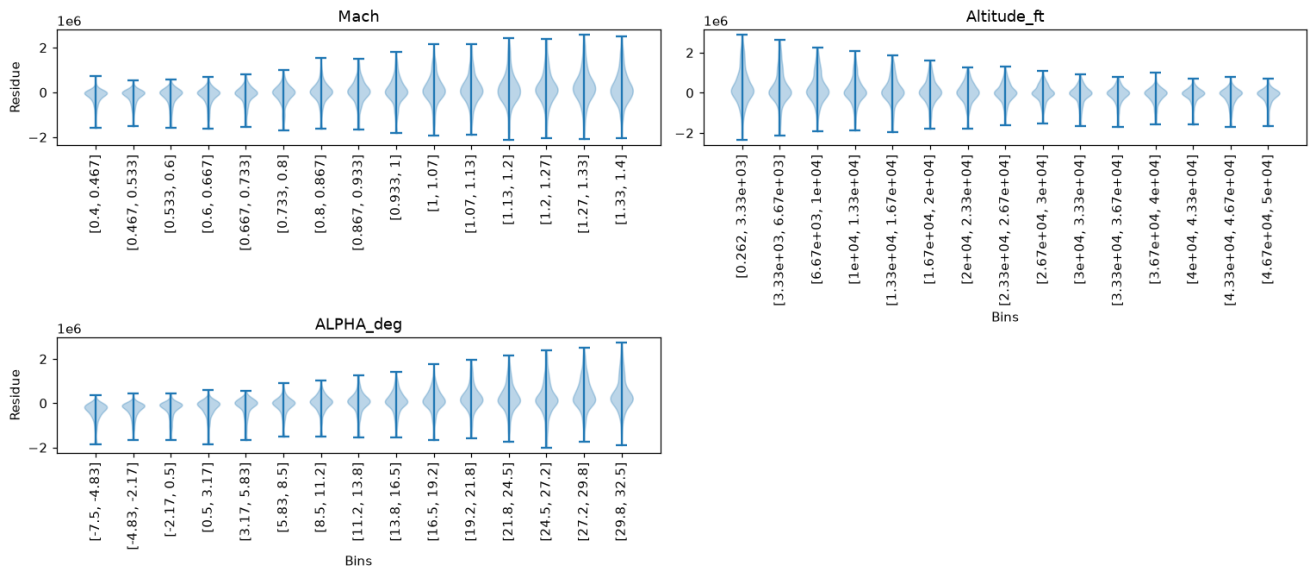
Residue — Fz_FUS

Fz_FUS
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$



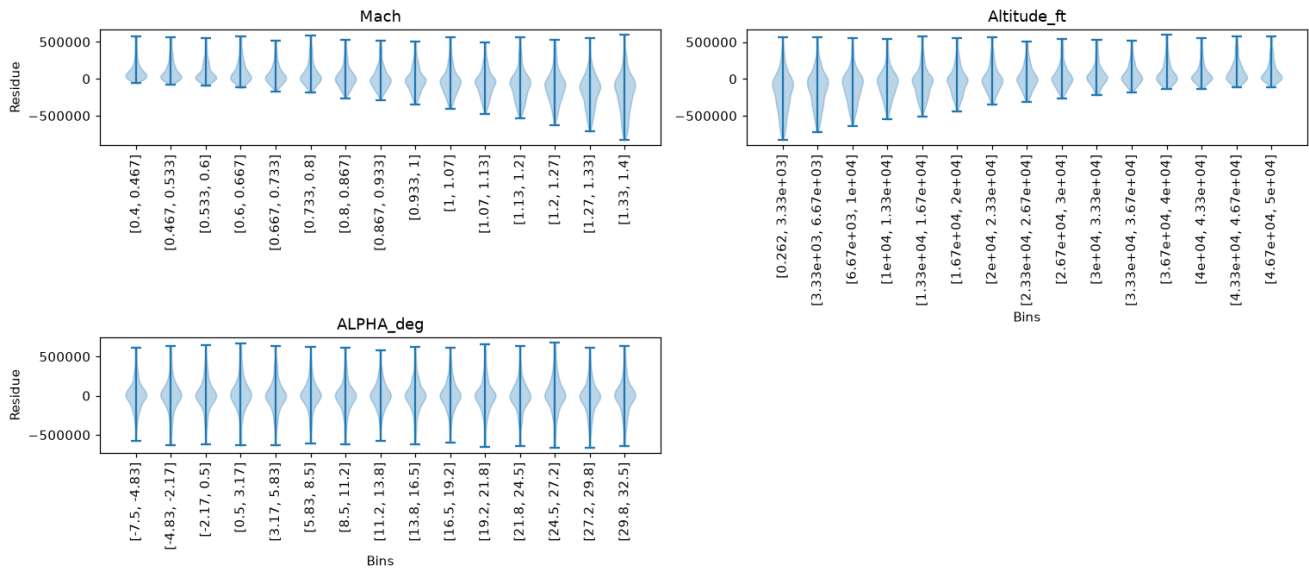
Residue — Mx_FUS

Mx_FUS
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$



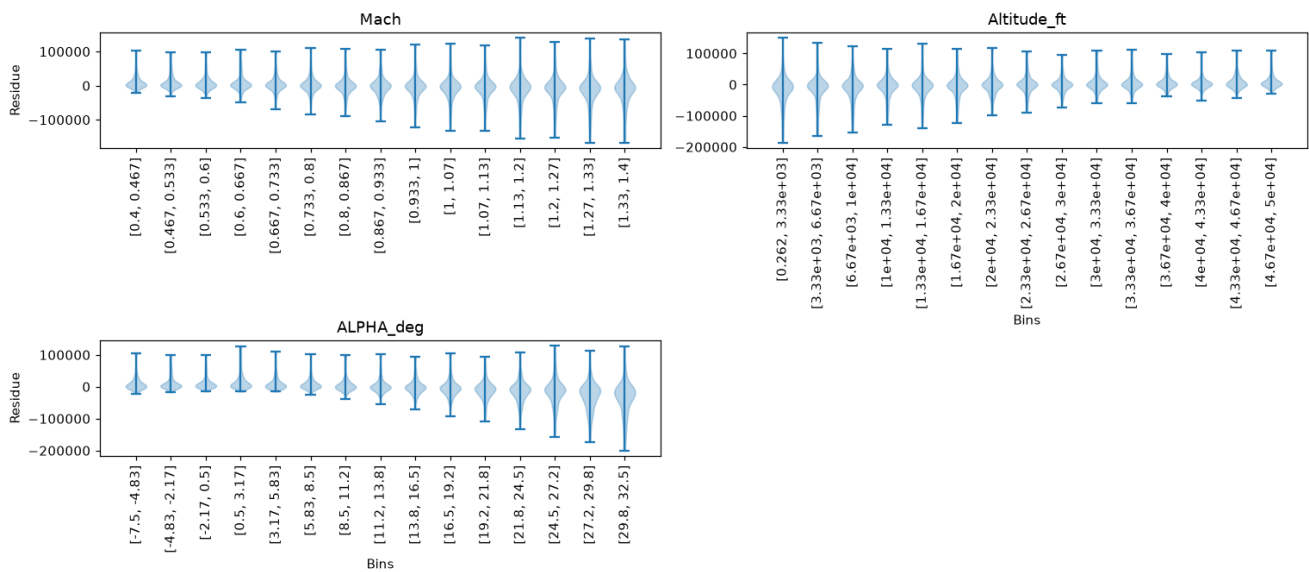
Residue — My_FUS

My_FUS
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$



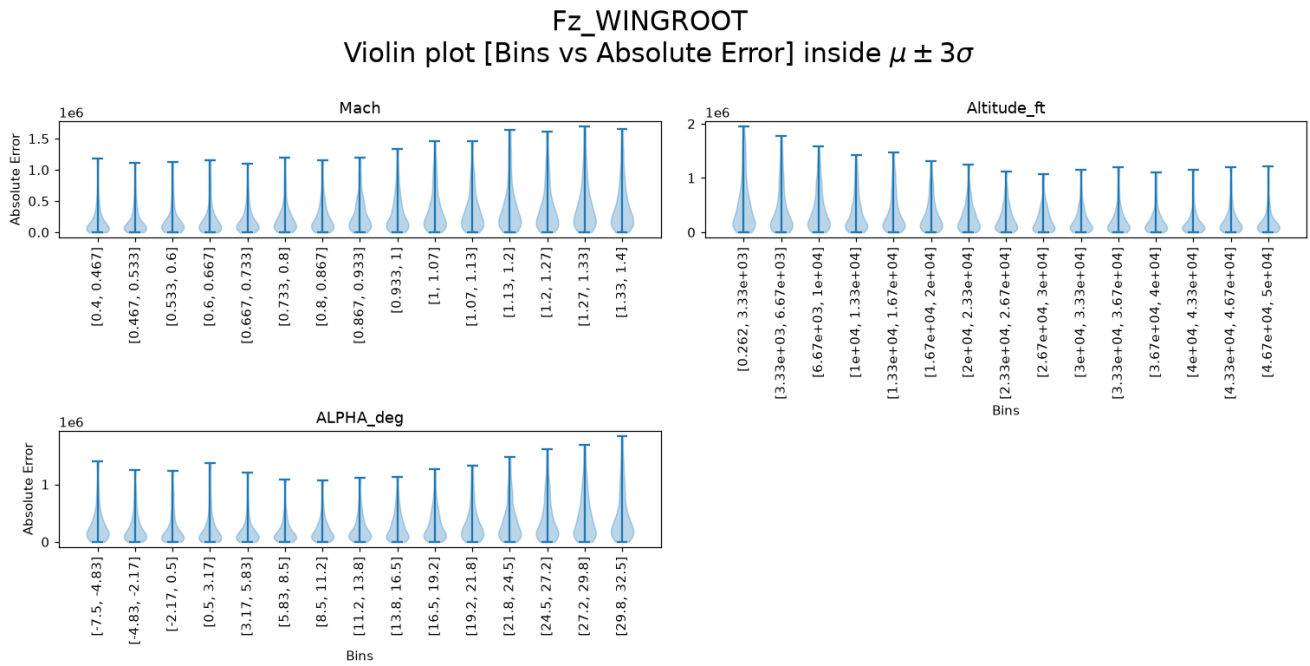
Residue — Mz_FUS

Mz_FUS
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$

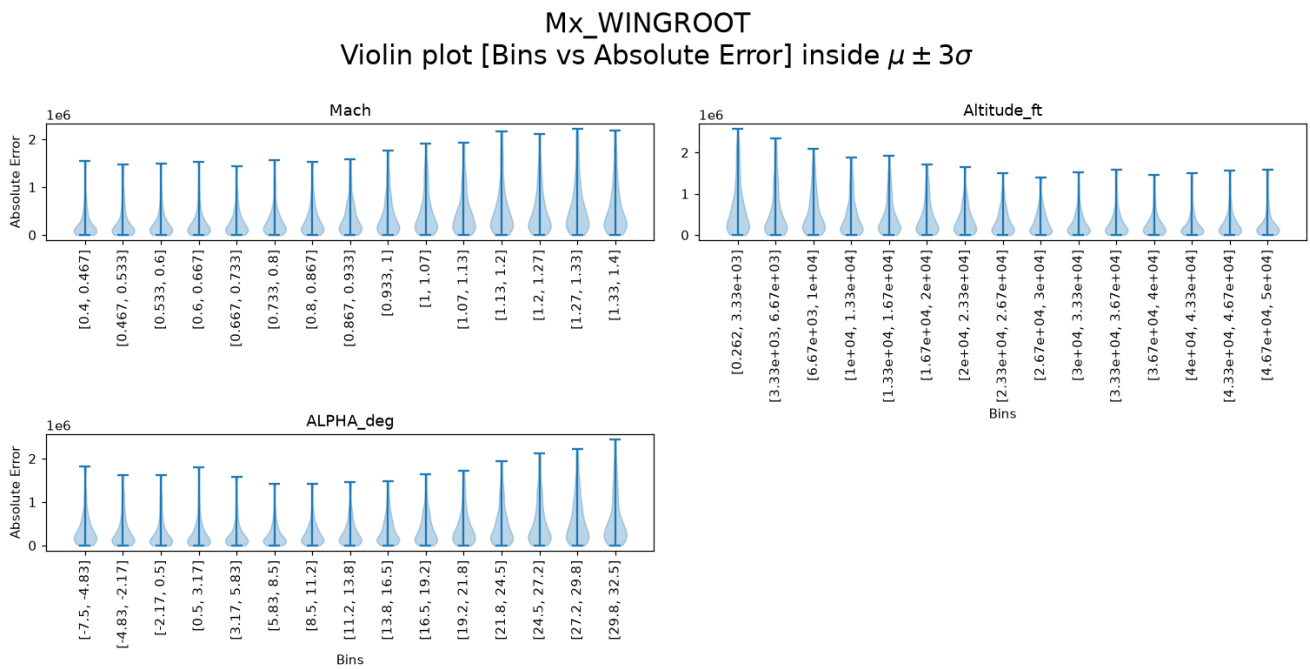


Absolute Error Violin Plots per Output (binned by input)

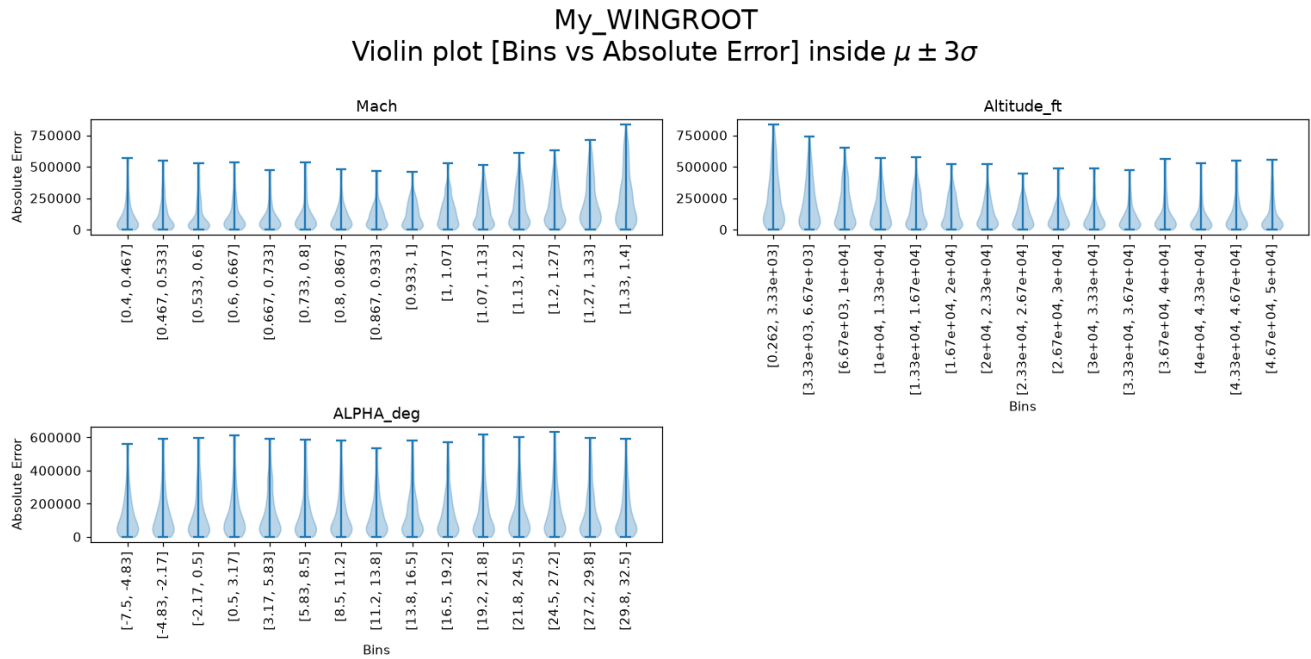
Absolute Error — Fz_WINGROOT



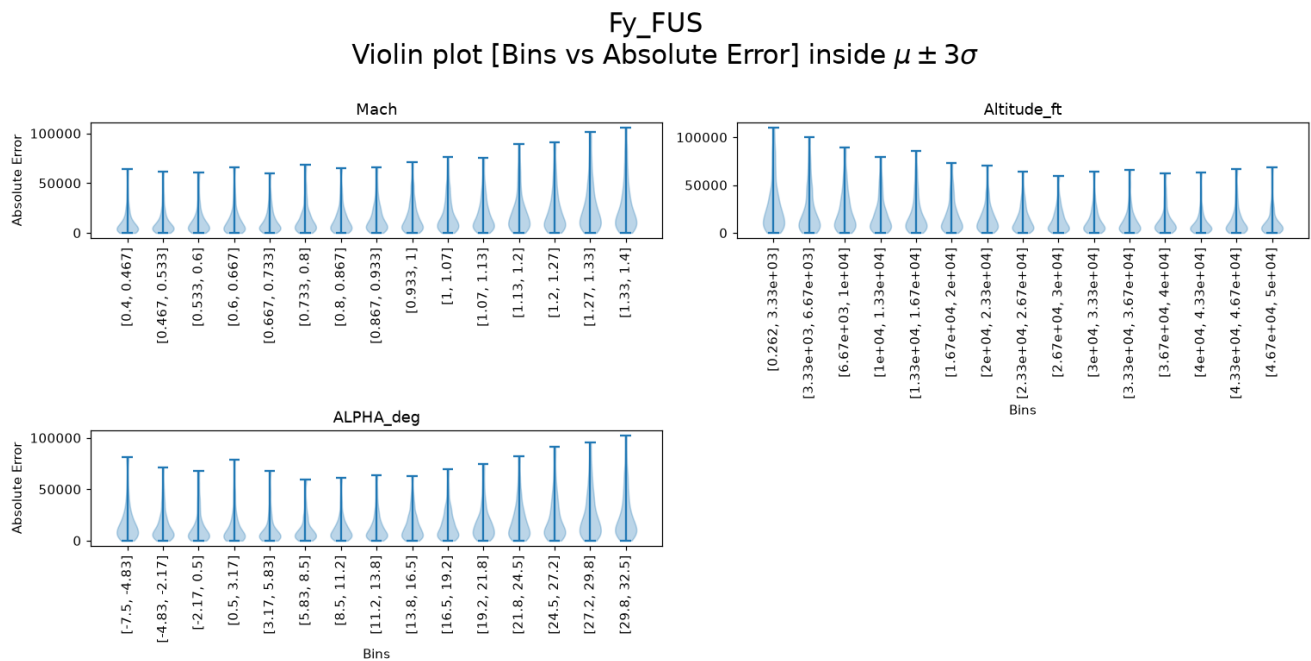
Absolute Error — Mx_WINGROOT



Absolute Error — My_WINGROOT

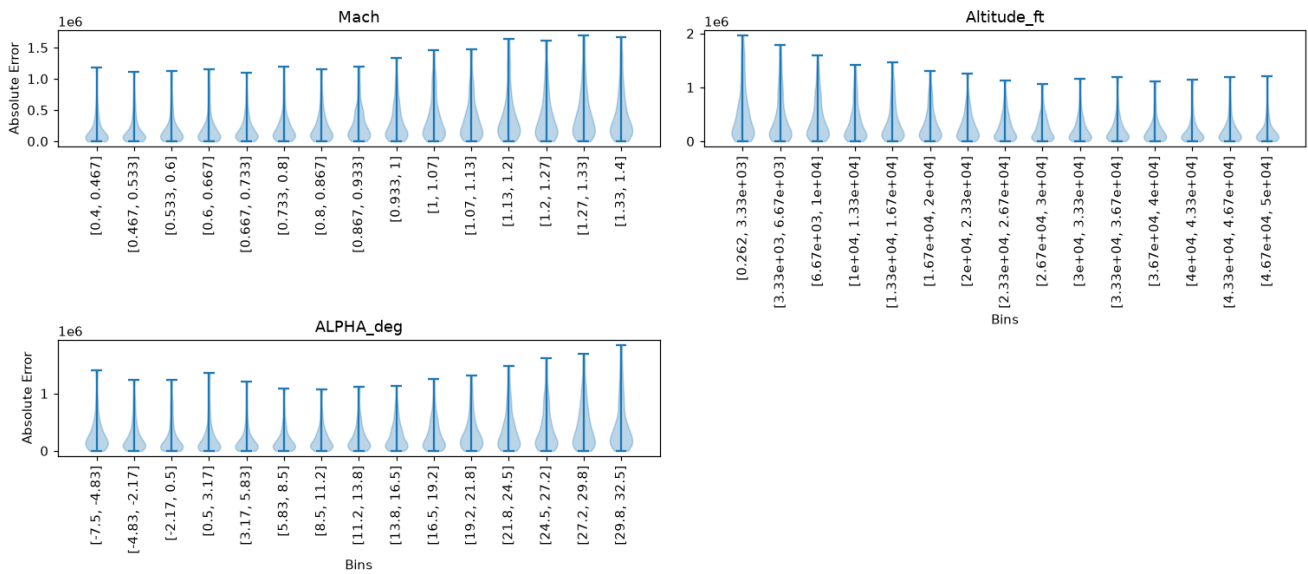


Absolute Error — Fy_FUS



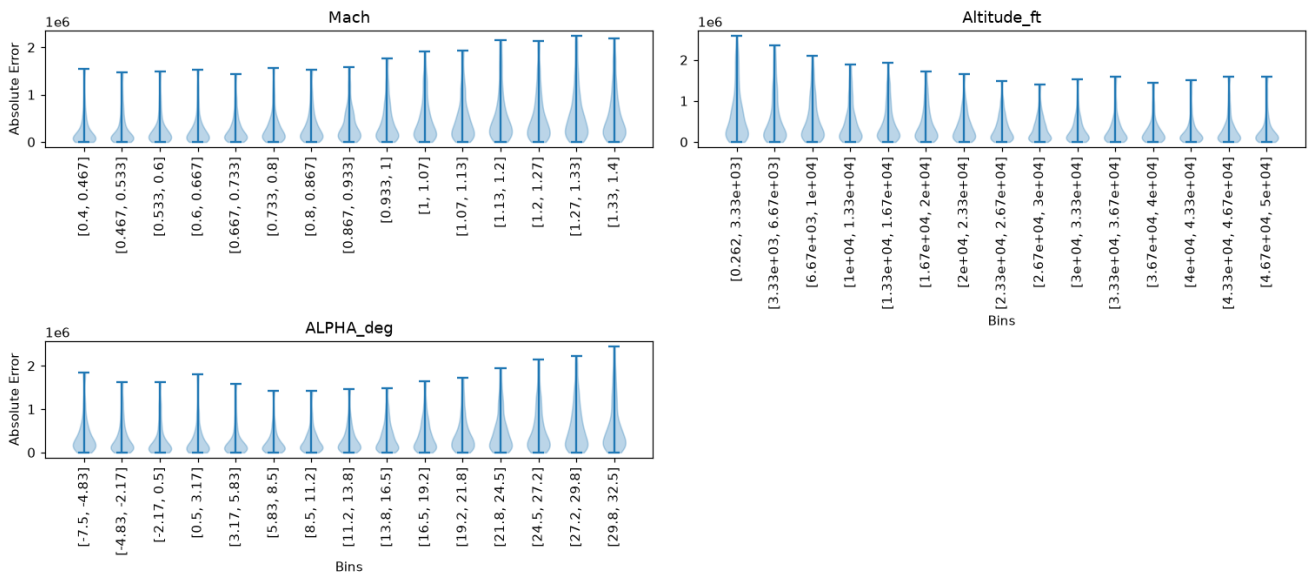
Absolute Error — Fz_FUS

Fz_FUS
Violin plot [Bins vs Absolute Error] inside $\mu \pm 3\sigma$

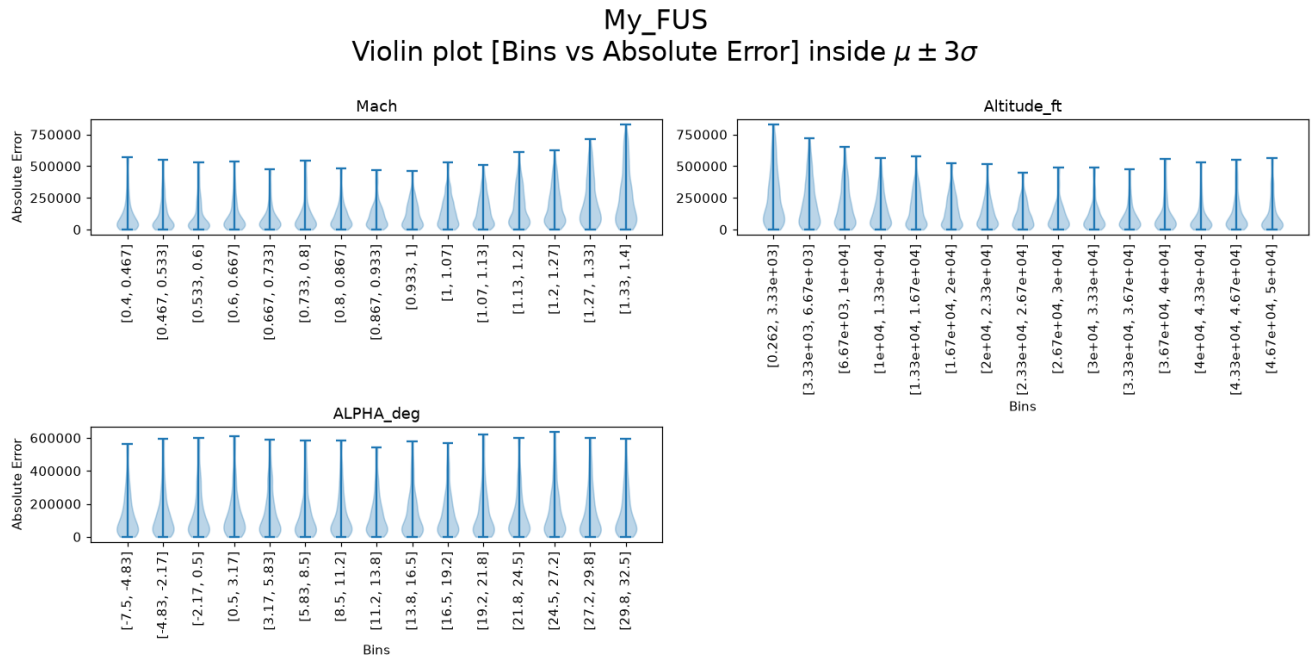


Absolute Error — Mx_FUS

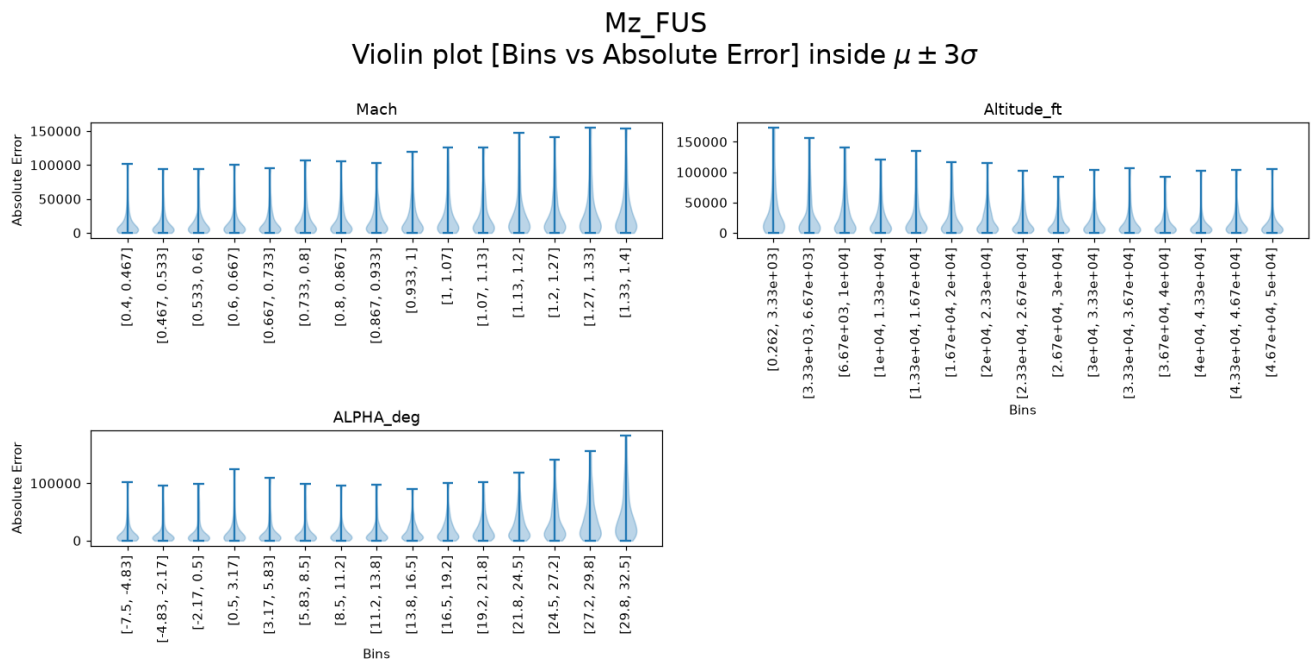
Mx_FUS
Violin plot [Bins vs Absolute Error] inside $\mu \pm 3\sigma$



Absolute Error — My_FUS



Absolute Error — Mz_FUS



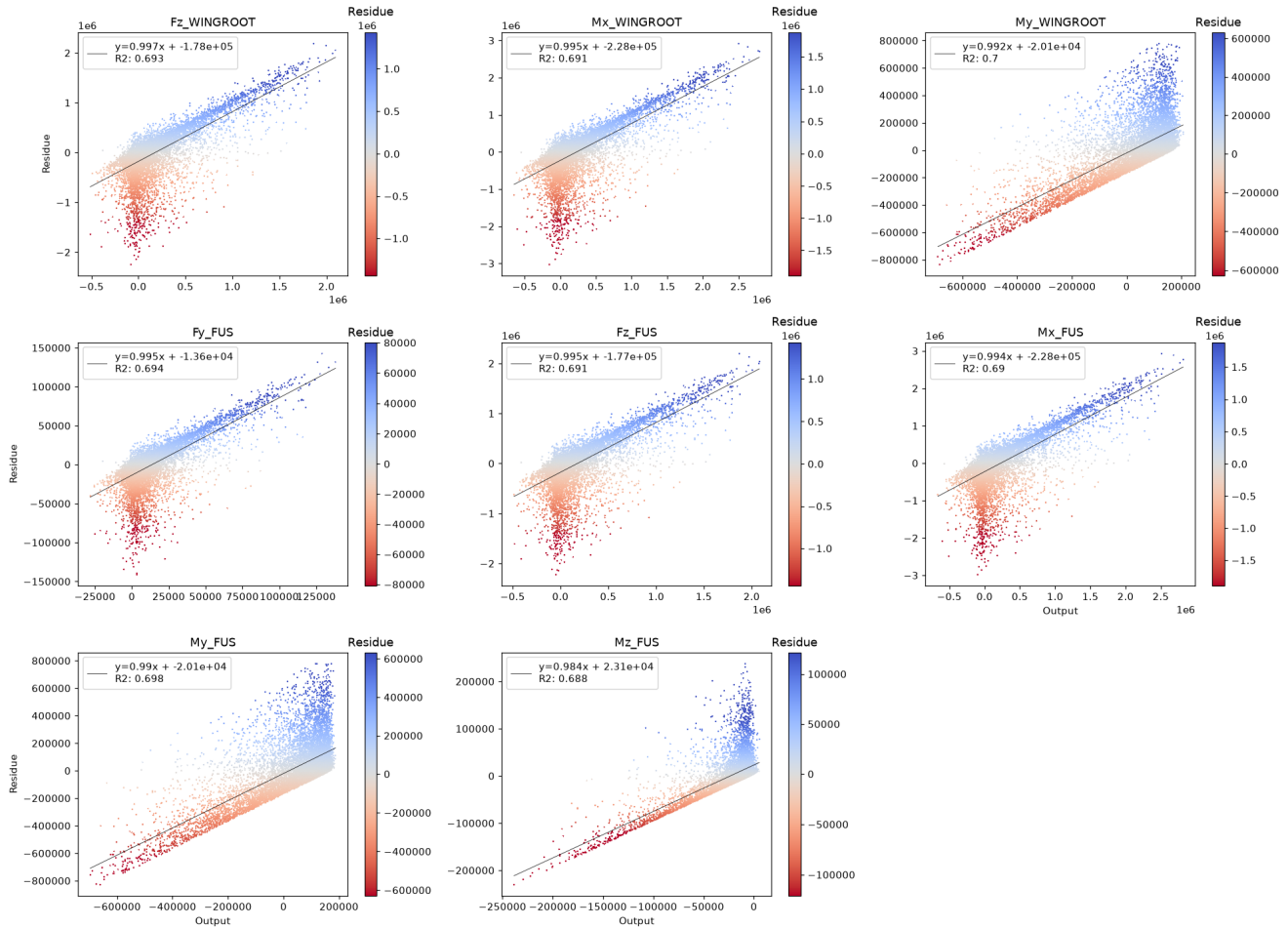
P(E|Y) — Error Conditional on Outputs

We study whether the error depends on the output magnitude. A significant trend (Pearson or Spearman $p < 0.05$) means the model is more accurate in some output ranges than others — a form of heteroscedasticity.

Residue vs True Output — Scatter

Pearson r and p-value shown in title. Red title = significant linear trend.

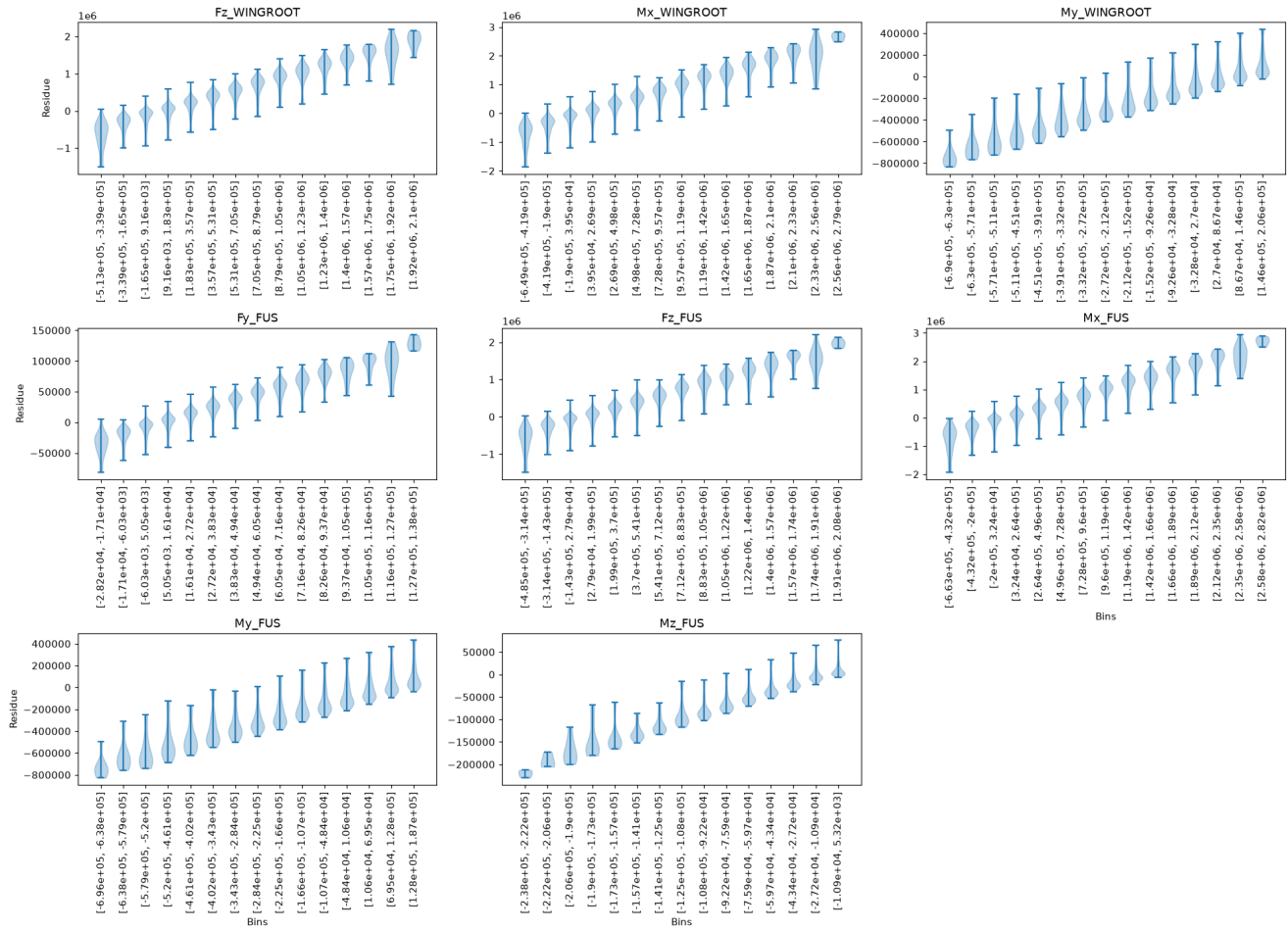
Scatter plot [Output vs Residue] (color trimmed to $\mu \pm 3\sigma$)



Residue Violin vs True Output Bins

Bins determined by Sturges rule. Median line dashed at 0.

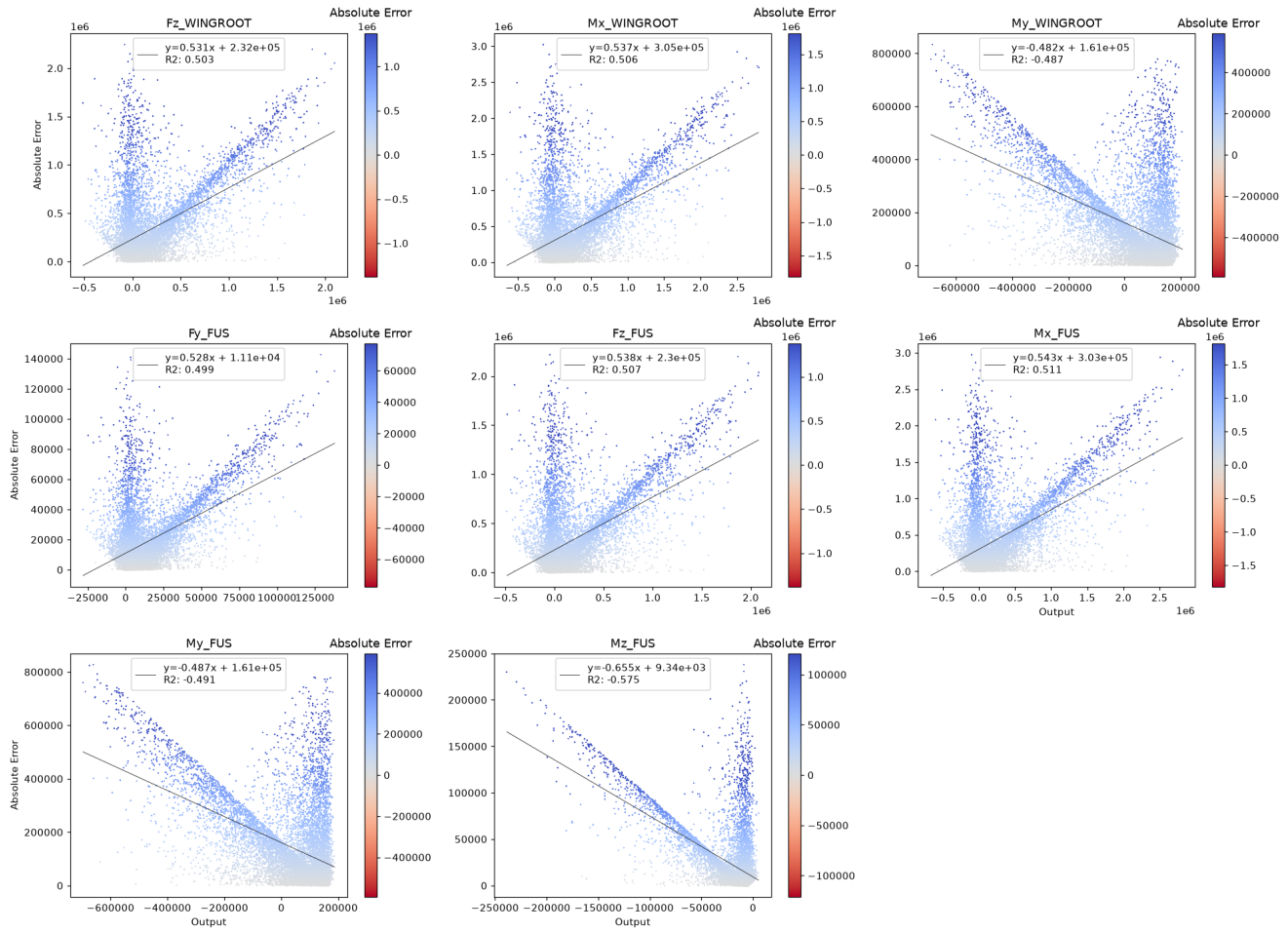
Violin plot [Bins vs Residue] inside $\mu \pm 2\sigma$



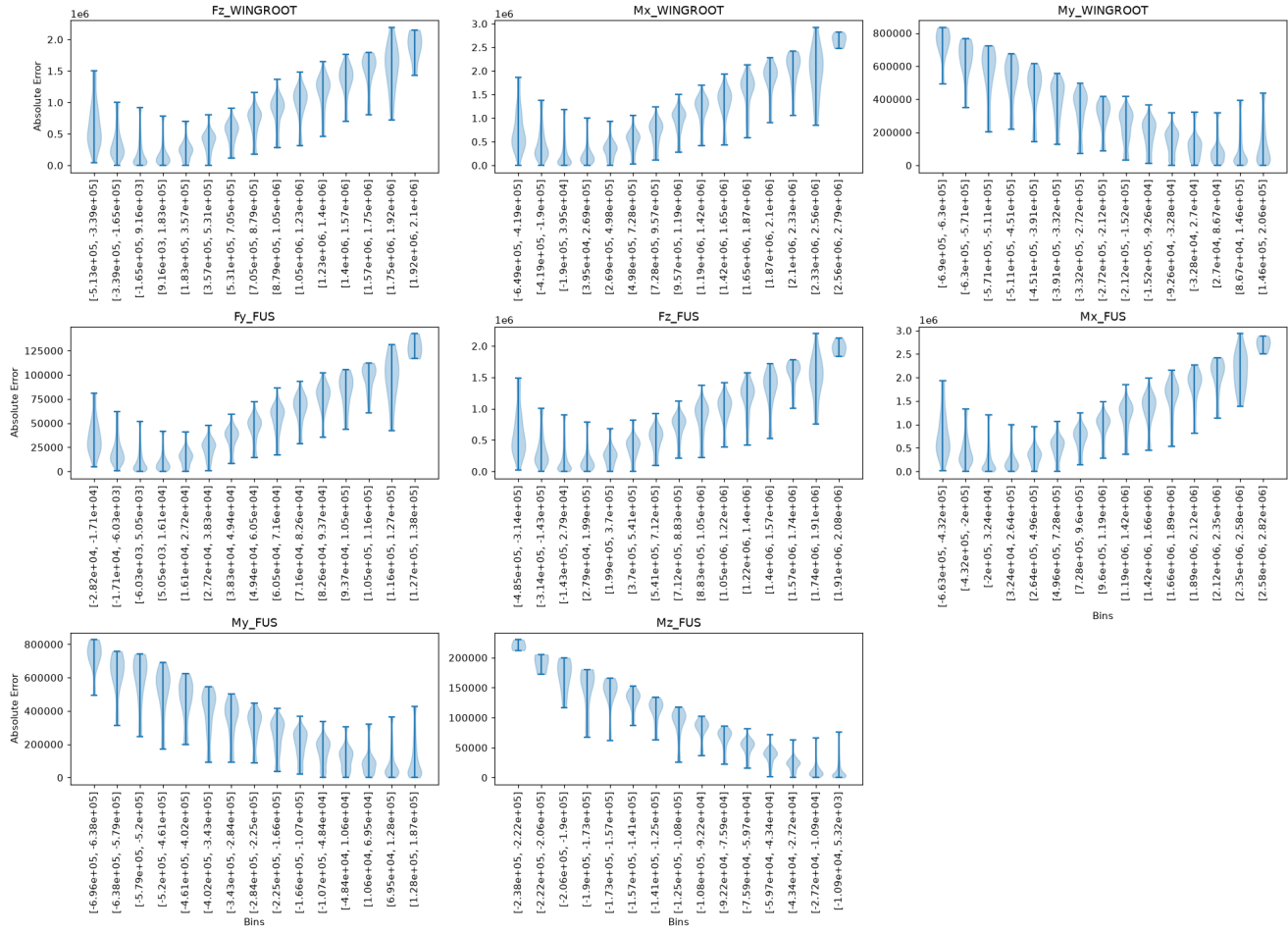
Absolute Error vs True Output — Scatter

Spearman r shown. Red title = significant monotonic trend.

Scatter plot [Output vs Absolute Error] (color trimmed to $\mu \pm 3\sigma$)



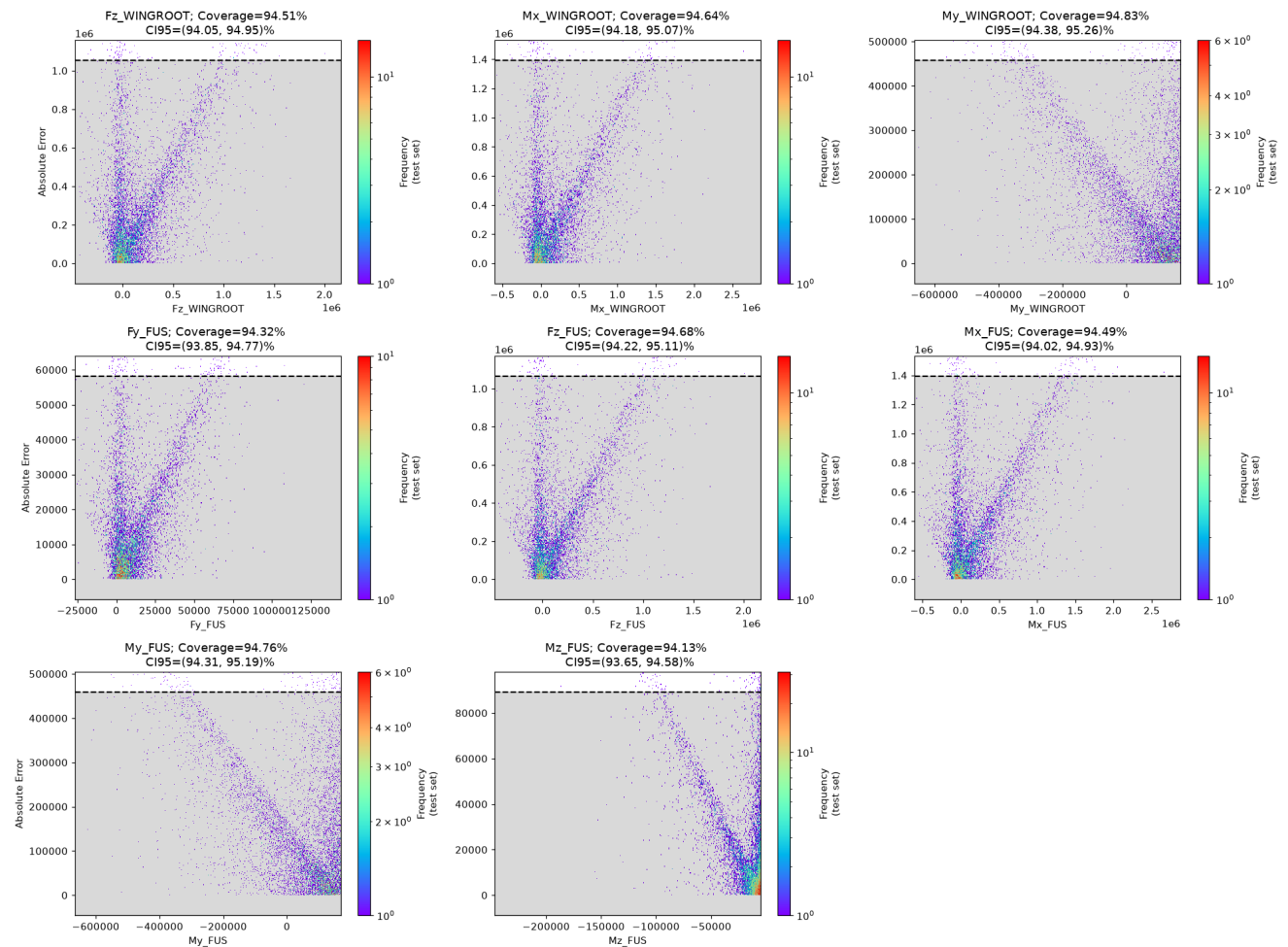
Absolute Error Violin vs True Output Bins

Violin plot [Bins vs Absolute Error] inside $\mu \pm 2\sigma$ 

Uncertainty Analysis

A global binned uncertainty model (95th percentile, right-tailed) is trained on the validation split of the absolute error and evaluated on the held-out test split, as in the extended validation notebook. Coverage is the proportion of test points whose error falls inside the predicted bound; it should be close to 95%.

Coverage Plot



Coverage Table

	Fz_WINGROOT	Mx_WINGROOT	My_WINGROOT	Fy_FUS	Fz_FUS	Mx_FUS	My_FUS	Mz_FUS	TOTAL COV
Coverage	94.51	94.64	94.83	94.32	94.68	94.49	94.76	94.13	—
CI	[94.05, 94.95]	[94.18, 95.07]	[94.38, 95.26]	[93.85, 94.77]	[94.22, 95.11]	[94.02, 94.93]	[94.31, 95.19]	[93.65, 94.58]	—

Empirical coverage of the 95th-percentile uncertainty bound on the test split.